Windowed Hankel LWE: A Structured-LWE Problem for Cryptanalysis, with a Compact KEM

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Abstract

Middle-Product LWE keeps the long polynomial operand secret and publishes many short operands. This arrangement admits reductions from Ring-LWE, but it is also why MP-LWE schemes are large, with public keys of about 16 KB. We study the dual arrangement, *Windowed Hankel LWE*. Its public matrix is a Hankel (equivalently, row-reversed Toeplitz) matrix generated by one long pseudorandom sequence, its secret is short, and only the convolution coefficients that depend on every secret coordinate are published: a noisy *middle product*. The problem sits between unstructured LWE and Ring- or Module-LWE, and needs no reduction modulo a fixed polynomial. On it we build a KEM with $n = 640$, $q = 2^{12}$ and binomial noise. A complete exchange takes 2,528 bytes (excluding format headers), the decryption failure rate is below $2^{-134.3}$, and a constant-time C implementation completes a handshake in 57 μs on an Apple M3 Max. We prove IND-CPA security in the random-oracle model under the hardness of the problem, and bound the IND-CCA security of the Fujisaki–Okamoto transform that the scheme uses. Treated as unstructured LWE, the cheapest generic attack costs $2^{137.1}$ in the Core-SVP model, against $2^{115.5}$ for Kyber-512. For the natural structural attacks we find no advantage over uniform matrices, under the models and experiments we examine: the embedding into the cyclic ring $\mathbb{Z}_q[X]/(X^{1407} - 1)$ and its subrings, shifted and biased dual functionals, algebraic decoding, low-rank Hankel methods, and lattice reduction at reduced dimensions. Because the MP-LWE reductions do not transfer, the hardness of Windowed Hankel LWE is a new assumption. We state what would count as a break, and we publish challenge instances, each with a uniform-matrix twin, to invite its cryptanalysis.

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Keywords: post-quantum cryptography, learning with errors, structured LWE, middle-product LWE, Hankel matrices, cryptanalysis, key encapsulation.

1 Introduction

Lattice-based key encapsulation mechanisms (KEMs) trade algebraic structure for efficiency. Unstructured LWE [37], as used by FrodoKEM [13], rests on worst-case reductions from standard lattice problems but needs about 19 KB per exchange at the 128-bit level. Module-LWE, as used by ML-KEM [14, 32], works in the ring $\mathbb{Z}_q[X]/(X^n + 1)$ and needs about 1.6 KB, at the price of a rich algebraic structure whose security has been studied extensively [34].

This paper studies an intermediate point. The public matrix is a *Hankel* matrix $H_{i,j} = a_{i+j}$ generated by a single pseudorandom sequence a, so that multiplying by it computes a contiguous slice of the coefficients of a polynomial product in $\mathbb{Z}_q[X]$, with no reduction modulo any polynomial. Which slice is published matters. The outermost coefficients of a linear convolution involve only

a few secret coordinates, and publishing them would expose the secret one coordinate at a time. We therefore publish only the *fully supported* coefficients, each of which involves all n secret coordinates. This is the *middle product* of Roşca, Sakzad, Steinfeld and Stehlé [38].

Their Middle-Product LWE problem (MP-LWE) uses the same operation, but places the long operand on the secret side and draws many short public operands. That arrangement is what makes reductions from Ring-LWE possible, and it is also a large part of why MP-LWE schemes are big: the Titanium KEM [39, 40] has a 16 KB public key. We invert the arrangement, publishing one long sequence and keeping the secret short, which brings a complete exchange down to 2,528 bytes. In exchange, we have no reduction. The security of the scheme rests on a new assumption, the hardness of the *Windowed Hankel LWE* problem, which we support with a detailed comparison to MP-LWE, an analysis of how known structure-exploiting attacks fare against it, and concrete estimates for every generic attack family.

The problem, not the scheme, is the main object of this paper. We offer Windowed Hankel LWE as a test case for cryptanalysis: an intermediate structured problem to which neither the reductions nor the structural attacks known on either side obviously apply. The KEM shows what the problem buys if it is hard, and gives attacks a concrete target; it is not meant for use until the problem has been analyzed independently.

Contributions.

1. A precise definition of Windowed Hankel LWE and an explicit hardness assumption (Section 4): how the problem differs from MP-LWE, why published structure-exploiting attacks do not apply as published, an embedding into a cyclic ring, and the effect of the power-of-two modulus. Table 2 summarizes the status of every attack family we considered, with what an adaptation would need, and Section 4.4 states what would count as a break. Experiments at reduced dimensions find no difference between Hankel and uniform instances, under lattice reduction or algebraic solving (Appendix E).

2. A KEM built on windowed Hankel products, in which a 128-coordinate window carries a 256-bit message by 4-PAM modulation and the modulus $q = 2^{12}$ reduces all arithmetic to masking (Section 5), with a game-based proof of IND-CPA security under Assumption 1 that accounts for the seeded public sequence, and IND-CCA bounds for the Fujisaki–Okamoto transform that the scheme implements, in the random-oracle and the quantum random-oracle models.

3. A parameter study, concrete estimates for all generic attack families, and an exact Chernoff bound on the decryption failure rate (Sections 6 and 7), with candidate parameters for NIST Categories 3 and 5 (Appendix D).

4. A constant-time C implementation that is bit-exact with the reference implementation, with a static audit of its compiled code and statistical timing tests (Section 8). The implementation, known-answer tests, parameter files, the scripts and raw outputs behind every figure, and challenge instances of the problem (Section 4.4) are available at `https://github.com/Cryptozoa/windowed-hankel-kem`.

After reviewing related work (Section 2) and fixing notation (Section 3), we define the problem (Section 4), specify the scheme and its parameters (Sections 5 and 6), analyze its security (Section 7) and describe the implementation (Section 8). Section 9 closes with limitations and open questions.

2 Related Work

Structured LWE. Ring-, Module- and Polynomial-LWE gain efficiency by working in polynomial rings [34]. Peikert and Pepin cast all of these, together with Order-LWE and MP-LWE, as

instances of a single generalized LWE problem defined by a bilinear form, and give simpler and tighter reductions among them [35]. In their framework, the matrix slices of the MP-LWE bilinear form are the standard basis of the Hankel matrices [35, Sect. 5], which is the sense in which our scheme is Hankel-structured. A separate line studies LWE for structured public matrices directly as a hardness question. Succinct LWE [42] and decomposed LWE [2] use matrices built from uniform blocks times short blocks plus a gadget matrix, and Abram et al. give both worst-case hardness and show them equivalent [1]. Their hardness comes from that decomposition, which Hankel matrices do not have, so these results do not transfer to our setting. Closer in form is the Toeplitz variant of LPN that Gilbert, Robshaw and Seurin introduced to shrink the keys of HB$^{\#}$, but it uses the arrangement of MP-LWE rather than ours: its secret is a Toeplitz matrix, which multiplies short public vectors, and its hardness is a conjecture [22].

Middle-Product LWE. Roşca et al. introduced MP-LWE and reduced Polynomial-LWE to it for every defining polynomial in a large family, so that MP-LWE is at least as hard as the hardest of these Polynomial-LWE instances [38]. Steinfeld, Sakzad and Zhao tightened the reduction and used it to set the parameters of Titanium [40], a first-round candidate in the NIST process that did not advance to the second round [3]. Peikert and Pepin reduce Ring-LWE to MP-LWE through Order-LWE with less error growth [35], and three recent works widen the coverage further: to at least 90% of bounded-coefficient polynomials of a given degree, experimentally [33]; to all number fields defined by monic irreducible polynomials with polynomially bounded coefficients [36]; and with less noise growth for trace-based reductions [17]. Bai et al. prove hardness for secrets that are small relative to the modulus, which their signature scheme MPSign requires [9], and derandomize the MP-LWE encryption scheme with a middle-product variant of learning with rounding that is at least as hard as decision MP-LWE [8]. Yang and Huang generalize the middle product to a universal product with more flexible parameters [43]. MP-LWE also supports identity-based encryption [28]; Tezuka and Tanaka survey these applications [41].

Cryptanalysis. Three published attacks exploit structure of the kind considered here: a dimension reduction for MP-LWE by Steinfeld et al. [40], a coefficient-by-coefficient recovery of small secrets by Bai et al. [9], and evaluation attacks on Ring-LWE over rings whose defining polynomial has a root of small order modulo q, by Elias, Lauter, Ozman and Stange [20]. Section 4.3 explains why none of them applies directly. Compact-LWE is a reminder that compact variants deserve suspicion: it claimed 138-bit security in dimension 13 from its departures from standard LWE, and generic lattice reduction decrypted its ciphertexts from the public key alone in a fraction of a second [12]. We take the opposite stance. The dimension is set by generic estimates, and the Hankel structure is treated as a potential weakness to be tested, not as a source of hardness. For concrete estimates we follow the Core-SVP methodology [6] and the analysis of Albrecht, Player and Scott [5], as implemented in the lattice estimator [4], and also report the cost model of MATZOV [30].

We found no published analysis of the setting studied here, with one long public sequence, a short binomial secret and narrow noise.

3 Preliminaries

Throughout, $q = 4096$ and $\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$, with elements represented in $[0, q)$, and vectors are indexed from 0. SHAKE256(x, ℓ) denotes ℓ bytes of SHAKE256 output on input x [31], $\|$ denotes concatenation, and $x[i{:}j]$ denotes bytes i to $j-1$ of a byte string x. The centered binomial distribution CBD_η outputs $\sum_{i<\eta}(x_i - y_i)$ for independent uniform bits x_i, y_i; in particular, CBD_2 takes the values $-2, \ldots, 2$ with probabilities $(1, 4, 6, 4, 1)/16$ and has variance 1. The scheme samples one CBD_2 value from the low four bits of each byte of SHAKE256 output.

3.1 Hankel matrices and the middle product

Definition 1 (Hankel matrix)**.** For $a = (a_0, \dots, a_{m+n-2}) \in \mathbb{Z}_q^{m+n-1}$, the $m \times n$ *Hankel matrix* of a is $H_m(a) \in \mathbb{Z}_q^{m \times n}$ with $H_m(a)_{i,j} = a_{i+j}$.

Write $a(X) = \sum_t a_t X^t$, and let $\tilde{s}(X) = \sum_{j<n} s_j X^{n-1-j}$ be the reversal of $s \in \mathbb{Z}_q^n$. Then

$$(H_m(a)\, s)_i = \sum_{j=0}^{n-1} a_{i+j}\, s_j = [X^{i+n-1}]\; a(X)\, \tilde{s}(X).$$

Of the $m + 2n - 2$ coefficients of $a(X)\tilde{s}(X)$, the vector $H_m(a)\, s$ keeps the m in the middle, which are exactly those involving all n entries of s; this is the *m-middle product* of [38]. Unlike circulant or negacyclic matrices, Hankel matrices are not closed under multiplication and do not form a ring.

Because convolution is commutative, we may also write $H_m(a)\, s = T_m(s)\, a$, where $T_m(s) \in \mathbb{Z}_q^{m \times (m+n-1)}$ is the banded matrix with $T_m(s)_{i,t} = s_{t-i}$ for $0 \le t - i < n$ and 0 otherwise. The two forms correspond to the two ways of assigning the long and the short operand, which is the distinction that matters in Section 4.2.

3.2 4-PAM modulation

A 256-bit message is carried by $W = 128$ coordinates, two bits each. Symbol $\mu \in \{0, 1, 2, 3\}$ is encoded as $\mathrm{PAM}(\mu) = \mu \cdot q/4 = 1024\mu$, and symbol w of the message consists of bits $2(w \bmod 4)$ and $2(w \bmod 4) + 1$ of byte $\lfloor w/4 \rfloor$.

Proposition 1 (Decoder)**.** If $v = 1024\mu + e \bmod 4096$ with $-512 \le e \le 511$, then $((v + 512) \bmod 4096) \gg 10 = \mu$.

Proof. Since $v + 512 \equiv 1024\mu + (e + 512) \pmod{4096}$ with $0 \le e + 512 \le 1023$, we have $\lfloor ((v + 512) \bmod 4096)/1024 \rfloor = \mu$, and division by 1024 is a right shift by 10. □

3.3 KEMs and the Fujisaki–Okamoto transform

A KEM is secure if the shared key is indistinguishable from random even to an adversary with a decapsulation oracle (IND-CCA). The Fujisaki–Okamoto transform with implicit rejection builds such a KEM from a public-key encryption scheme that is IND-CPA secure and δ-correct, where δ is the probability that decryption of an honestly generated ciphertext fails. Hofheinz, Hövelmanns and Kiltz analyze it in the random oracle model [23], and Jiang et al. [24] and Bindel et al. [11] in the quantum random oracle model; in all of these bounds the adversary's advantage grows with δ.

4 The Windowed Hankel LWE Problem

4.1 Definition

Definition 2 (Windowed Hankel LWE)**.** Let $n, \kappa \ge 1$, $m_1, \dots, m_\kappa \ge 1$ and $q \ge 2$, and let χ_s and χ_e be distributions on $\mathbb{Z}$. In $\mathsf{WH\text{-}LWE}_{n,(m_1,\dots,m_\kappa),q,\chi_s,\chi_e}$, a secret $s \leftarrow \chi_s^n$ is sampled and, independently for each $\ell \le \kappa$, a uniform $a^{(\ell)} \in \mathbb{Z}_q^{n+m_\ell-1}$ and an error $e^{(\ell)} \leftarrow \chi_e^{m_\ell}$, which give $b^{(\ell)} = H_{m_\ell}(a^{(\ell)})\, s + e^{(\ell)} \bmod q$. The *search* problem is to find s from $(a^{(\ell)}, b^{(\ell)})_{\ell \le \kappa}$; the *decision* problem is to distinguish these pairs from pairs in which each $b^{(\ell)}$ is replaced by a uniform vector.

The scheme relies on two instances, both with $n = 640$, $q = 4096$ and $\chi_s = \chi_e = \mathrm{CBD}_2$: the public key is an instance with $\kappa = 1$ and $m_1 = 768$, and the ciphertext, once the public key has been replaced by a uniform vector in the security argument, is an instance with $\kappa = 2$ and $(m_1, m_2) = (768, 128)$ whose secret is the ephemeral r (Section 5.3). Read as plain LWE, each is

an instance with secret dimension n and $m_1 + \cdots + m_\kappa$ samples whose matrix happens to be Hankel.

Assumption 1 (Windowed Hankel LWE)**.** For $n = 640$, $q = 4096$, $\chi_s = \chi_e = \mathrm{CBD}_2$ and $(m_\ell) = (768)$ or $(768, 128)$, decision WH-LWE is as hard as decision LWE with the same secret dimension, number of samples, modulus and distributions, and a uniformly random matrix: no attack is significantly cheaper than the best generic attack on the latter.

The security of the KEM rests on this assumption and on nothing else beyond the standard random-oracle arguments (Section 5.3). It is the claim this paper asks cryptanalysts to test, and Section 4.4 states what would refute it.

4.2 Comparison with Middle-Product LWE

In MP-LWE [38, Def. 3.5], the secret is a uniform polynomial of degree less than $n + d - 1$, and the adversary receives arbitrarily many samples, each pairing a fresh uniform polynomial a_i of degree less than n with d noisy coefficients of the middle product of a_i and the secret. Roşca et al. observe that this is LWE with correlated samples, sample i multiplying the secret by a banded $d \times (n + d - 1)$ Toeplitz matrix built from a_i [38, Sect. 3.2]. In the language of Peikert and Pepin, sample i applies a $d \times n$ Hankel matrix generated by the *secret* to the coefficient vector of a_i [35, Def. 5.1]. Our problem is the mirror image: the Hankel matrix is generated by the *public* sequence and applied to the secret. Table 1 sets the two side by side.

	MP-LWE [35, 38]	Windowed Hankel LWE (this work)
Hankel matrix generated by	the secret (length $n + d - 1$)	the public sequence (length $n+m-1$)
Applied to	a fresh short public polynomial per sample	the short secret
Number of samples	arbitrarily many; $\Theta(\log n)$ in the encryption scheme of [38]	one (public key) or two (ciphertext)
Secret	uniform, or Gaussian of width growing with $\sqrt{n}$ [9]	CBD_2, standard deviation 1
Error	Gaussian of width $\Omega(\sqrt{n})$	CBD_2, standard deviation 1
Hardness	reductions from Ring-/Polynomial-LWE	none known

Table 1: MP-LWE and Windowed Hankel LWE.

The known reductions depend on the MP-LWE arrangement at every step. They embed a Polynomial-LWE secret into the long MP-LWE secret, rerandomize across many samples, and need noise wide enough to absorb the distortion of the embedding [33, 35, 36, 38]; Roşca et al. also choose the width $\Omega(\sqrt{n})$ explicitly to rule out attacks in the style of Arora and Ge [38, Sect. 4]. With a single public sequence, a short secret and noise of standard deviation 1, none of these steps is available to us. Commutativity does not repair this: $H_m(a)\, s = T_m(s)\, a$ is an identity between matrices, not between security games, since the noise, the entropy and the choice of hidden operand all sit on different sides.

What can be said is limited. Windowed Hankel LWE is plain LWE whose matrix happens to be Hankel, so every generic attack applies to it unchanged. That does not make it formally at most as hard as plain LWE: Hankel matrices are a vanishing fraction of all matrices, so guarantees proved for algorithms on uniform matrices do not transfer. At reduced dimensions, generic attacks behave the same on both (Appendix E), and generic estimates are the optimistic direction for the defender. The converse fails on a dimension count: a Hankel matrix has $n + m - 1 = 1{,}407$ free entries against $nm = 491{,}520$ for a uniform matrix, so a generic LWE instance cannot be

embedded in one. Few degrees of freedom are not a weakness in themselves. ML-KEM-512 determines its 512×512 matrix by 1,024 coefficients, fewer than our 1,407, while the matrix of FrodoKEM-640 has 409,600 independent entries; all three are expanded from seeds of 128 or 256 bits. What ML-KEM has and Windowed Hankel LWE lacks is a reduction explaining why its structure is harmless. Nor is the public key statistically hidden: the secret carries about 1,300 bits of entropy, against 1,195 for ML-KEM-512 and about 2,260 for one column of the secret of FrodoKEM-640, while b has $768 \times 12 = 9{,}216$ bits, so hiding is computational and depends on the structure. Finally, the cyclic restatement of Section 4.3 does not import hardness from cyclotomic Ring-LWE: the ring $\mathbb{Z}_q[X]/(X^L - 1)$ has the small-order root 1, and relating it to a cyclotomic ring would mean working with complete samples, which is what the withheld coefficients prevent. Windowed Hankel LWE thus lies off both branches of the structured-LWE landscape, neither unstructured nor inherited from a ring, and we treat its hardness as an assumption in its own right, checking resistance to Arora–Ge attacks concretely in Section 7.

4.3 Structure and known attacks

A Hankel matrix $H_m(a)$ is determined by $n + m - 1$ values rather than nm, and its rows are shifts of one another. We now ask whether known attacks can exploit this; Section 7 treats the generic attacks.

Full support defeats truncation and peeling. Since $b_i = \sum_{j<n} a_{i+j} s_j + e_i$, every published coordinate depends on all n secret coordinates: the structure of $H_m(a)$ forces none of its entries to be zero. This rules out two published techniques. Steinfeld et al. reduce the dimension of the primal attack on MP-LWE by discarding output coordinates, which removes secret coordinates because each output sees only a window of the long secret [40, Prop. 2]. Here, discarding outputs removes no secret coordinate; it merely selects a subset of samples, which the generic attacks already optimize. Bai et al. recover a small secret coefficient by coefficient from the lowest coefficients of a full, noiseless product, where the r-th coefficient involves only secret coefficients of index at most r [9, Sect. 4]. Our window excludes those coefficients, and our outputs are noisy.

Embedding into a cyclic ring. Let $L = n + m - 1$ be the length of the public sequence. Reducing $a(X)\tilde{s}(X)$, of degree at most $L + n - 2$, modulo $X^L - 1$ folds only the coefficients of degree at least L, onto degrees $0, \ldots, n - 2$. The published coefficients, of degrees $n - 1$ to $L - 1$, are untouched, so

$$b_i = [X^{i+n-1}] \left(a(X)\, \tilde{s}(X) \bmod (X^L - 1) \right) + e_i, \qquad 0 \le i < m.$$

A public key is therefore a Ring-LWE-type sample over $\mathbb{Z}_q[X]/(X^L - 1)$ whose secret is supported on n of the L coefficients and of which only m output coefficients are revealed. This ring is far from the cyclotomic rings recommended for Ring-LWE, since $X^L - 1$ has the root 1 modulo every q. Over such rings, evaluating a sample at a root α of the defining polynomial collapses it to a one-dimensional sample $b(\alpha) = a(\alpha)s(\alpha) + e(\alpha)$, and this is the basis of the attacks of Elias et al. [20].

The evaluation, however, needs all L coefficients of the sample, and $n - 1 = 639$ of them are withheld. With only the window, an attacker can apply a linear functional $\varphi \in \mathbb{Z}^m$ and obtain

$$\varphi^\top b = \left(H_m(a)^\top \varphi \right)^\top s + \varphi^\top e.$$

For a factor g of $X^L - 1$ over $\mathbb{Z}_q$ of degree k, let $R_g \in \mathbb{Z}_q^{k \times n}$ be the matrix of the map $s \mapsto \tilde{s} \bmod g$. Functionals with $H_m(a)^\top \varphi = R_g^\top c$ for some $c \in \mathbb{Z}_q^k$ give $\varphi^\top b = c^\top (\tilde{s} \bmod g) + \varphi^\top e$, which depends on s only through the k coefficients of $\tilde{s} \bmod g$; for $g = X - 1$ this is $s(1) = \sum_j s_j$. The all-ones functional does not qualify, because the entries $\sum_{i<m} a_{i+j}$ of $H_m(a)^\top \mathbf{1}$ vary with j: truncation

does not commute with evaluation. The qualifying functionals form a q-ary lattice Λ_g of rank m, cut out by $n-k$ linear constraints modulo q. To use them, an attacker needs vectors of Λ_g short enough that $\varphi^\top e$ is far from uniform, and must then guess $\tilde{s} \bmod g$.

This is the situation of the dual hybrid attack of Section 7, which likewise drops $n-k$ constraints and guesses k quantities, except that there the guessed quantities are k coordinates of s. The factors are known explicitly. Since L is odd, $X^L - 1 = \prod_{d|L} \Phi_d$ is squarefree modulo 2, so its factorization over $\mathbb{Z}_q$ lifts by Hensel's lemma from that over $\mathbb{F}_2$, where Φ_d splits into $\deg(\Phi_d)/\mathrm{ord}_d(2)$ irreducible factors of degree $\mathrm{ord}_d(2)$. For $1407 = 3 \cdot 7 \cdot 67$, the order of 2 is 2, 3 and 6 modulo 3, 7 and 21, and 66 modulo each of 67, 201, 469 and 1407, so $X^{1407} - 1$ has irreducible factors of degrees 1, 2, 3, 3, 6 and 6, and $1 + 2 + 6 + 12 = 21$ factors of degree 66. Appendix C reports two computations for the low-degree factors. At full size, Λ_g has determinant exactly q^{n-k} for every random sequence tested, the same as the corresponding dual-hybrid lattice, while the quantity to be guessed carries more entropy than k coordinates of s: 6.7 bits for $s(1)$ against 2.0 bits for one coordinate, and about 33 against 6.1 bits for each cubic factor. Since each factor g divides some Φ_d with $d \in \{1, 3, 7, 21\}$, the residue $\tilde{s} \bmod g$ is a function of $\tilde{s} \bmod \Phi_d$, whose coefficients are small integers, and its entropy is at most that of this short vector. Composite divisors do not help either: for $G = \Phi_7$ or $X^7 - 1$, the guess carries 33.2 or 37.1 bits, against 12.2 or 14.2 bits for as many coordinates of s (Appendix C). At reduced dimensions, where shortest vectors can be computed exactly, the lattices Λ_g built from Hankel matrices have shortest vectors within a few percent of the Gaussian heuristic, like lattices built from uniform matrices: over 315 instances of each kind, the mean ratio lies between 0.99 and 1.02 for every factor and size. Under these measures, functionals of this kind offer nothing that the generic dual hybrid attack does not already provide more cheaply. Equal determinants do not by themselves make two lattice problems equally hard, which is why we also compare shortest vectors. The same analysis applies to any binomial $X^L - \alpha$ with α a unit: reducing modulo it also folds only the coefficients of degree at least L, onto the withheld ones, and modulo 2 it coincides with $X^L - 1$, so its factors have the same degrees; for $\alpha = -1$, the factor $X + 1$ isolates $\tilde{s}(-1)$, with the same entropy as $s(1)$. Switching to a prime modulus q' that splits $X^d - 1$ into linear factors does not avoid the withheld coefficients either: the functionals still need $n-k$ constraints, now modulo q', and rounding a to the new modulus adds noise of standard deviation about $\|s\|/\sqrt{12} \approx 7.3$, against 1 for e. This does not exclude other uses of the ring structure, which remain open.

Biased functionals. A functional need not isolate $\tilde{s} \bmod g$ exactly. If $H_m(a)^\top \varphi = R_g^\top c + w$ with w short, then $\varphi^\top b = c^\top (\tilde{s} \bmod g) + w^\top s + \varphi^\top e$ still correlates with $\tilde{s} \bmod g$. The pairs (φ, w) form a lattice of dimension $m + n$ and determinant q^{n-k}; for $k = 0$ it is the lattice of the dual attack. The dual hybrid attack that guesses k coordinates of s works with the same determinant in dimension $m + n - k$, so the biased lattice is slightly denser: at full size, its Gaussian heuristic is lower by at most 0.02 bits, for the sextic factors, so its shortest vectors are at most about 1.4% shorter. The guess, however, is again $\tilde{s} \bmod g$, which costs 4.7 to 49.5 bits more than k coordinates of s (Table 10). At reduced sizes, after removing the trivial vectors $(0, R_g^\top c)$, which carry no information, Hankel and uniform matrices give biased lattices with the same shortest vectors (Appendix C).

Subrings with the withheld coefficients as unknowns. A primal or bounded-distance-decoding attack could instead work in a subring $\mathbb{Z}_q[X]/(G)$ and treat the $n-1$ withheld coefficients as unknowns. For linear attacks of this kind, the following proposition shows that the subrings give no information beyond the window equations.

Proposition 2. Let $c = a(X)\, \tilde{s}(X) \bmod (X^L - 1)$, let $u = (c_0, \ldots, c_{n-2})$ be its withheld coefficients, and let G be a monic divisor of $X^L - 1$ over $\mathbb{Z}_q$ of degree k. If $k \leq n - 1$, then $c \bmod G$ takes every value in $\mathbb{Z}_q[X]/(G)$ as u varies, with the window fixed. In general, every

$\mathbb{Z}_q$-linear combination of the coefficients of the relation $c \bmod G = (a \bmod G)(\tilde{s} \bmod G)$ that does not involve u is a linear combination of the window equations $b - e = H_m(a)\, s$.

Proof. Write $c = u(X) + w(X)$, where $u(X) = \sum_{i<n-1} c_i X^i$ and $w(X)$ collects the window. Reduction modulo G is linear, so $c \bmod G = P_G u + (w \bmod G)$, where column i of P_G holds the coefficients of $X^i \bmod G$. For $k \le n-1$, the columns $i < k$ form the identity, so P_G is onto. In general, a combination μ that does not involve u satisfies $\mu^\top P_G = 0$, and the equation it yields reads $\mu^\top (w \bmod G) = \mu^\top \big((a \bmod G)(\tilde{s} \bmod G) \bmod G\big)$. Both sides are linear in s once the window is written as $w = H_m(a)\, s$, and the relation holds identically in s, so the equation is $\mu^\top Q_G \big(w - H_m(a)\, s\big) = 0$, where Q_G reduces window coefficients modulo G: a linear combination of the window equations, with $w = b - e$. □

Every irreducible factor of $X^{1407} - 1$ has degree at most 66, far below $n - 1 = 639$, so each subring equation on its own carries no information about s, and the evidence directory confirms this for all 27 factors. A divisor of degree $k \ge n$ leaves $\tilde{s}$ unreduced, and its u-free equations are linear combinations of the window equations, which the generic attacks of Section 7 already use directly. An attack that keeps u as unknowns instead has k equations, of which eliminating the $n-1$ withheld coefficients leaves $k-n+1$: the same as using $k-n+1$ window equations directly. Eliminating u from the full relation leaves exactly $b = H_m(a)\, s + e$. The withheld coefficients are not free variables, but noiseless linear functions of s and a; substituting that dependence makes their terms cancel identically, which is why the relation carries no information beyond the window. Attacks that would escape the proposition must be non-linear in the ring, through norms or the automorphisms $X \mapsto X^t$, for instance, and these act on the full coefficient vector, including u.

Shifted functionals. In a ring lattice, multiplying a short vector by X gives another short vector of the same lattice. Hankel matrices have a weaker analogue. If $\varphi \in \mathbb{Z}^m$ satisfies $(H_m(a)^\top \varphi)_j \equiv 0$ for $j \in J$ and $\varphi_{m-1} = 0$, then its shift $(S\varphi)_i = \varphi_{i-1}$, with $(S\varphi)_0 = 0$, has the same length and satisfies the constraints on $J - 1 = \{j - 1 : j \in J,\ j \ge 1\}$, since $(H_m(a)^\top S\varphi)_j = \sum_{i \ge 1} a_{i+j}\varphi_{i-1} = (H_m(a)^\top \varphi)_{j+1}$ for $j \le n-2$; an attacker obtains $\varphi_{m-1} = 0$ by leaving out one sample. Unlike a ring rotation, $S\varphi$ leaves the lattice: on 399 BKZ-reduced vectors at reduced size, the shifted constraints held every time and the dropped constraint never did. A shifted functional is therefore useful only to a dual hybrid attack that guesses one more coordinate of s, and that attack can instead reduce the lattice with one constraint fewer, whose determinant is smaller by a factor of q. Forcing r consecutive shifts of one functional into the same lattice requires its constraints on $J, J-1, \ldots, J-r+1$ at once, that is $n-k+r-1$ constraints on vectors with $r-1$ fixed zeros: a determinant larger by q^{r-1} in $r-1$ fewer dimensions, so the vectors are longer. Shifts can supply more vectors, but not shorter ones. Granting the dual hybrid attack a free 640-fold supply of short vectors, with no extra guessing, lowers its estimate by only 0.9 bits under MATZOV and 1.0 bit under Core-SVP: at the optimum, a single sieve call already yields the $2^{94.9}$ vectors that the distinguisher needs.

Algebraic decoding. Without noise, $b = H_m(a)\, s$ is a Hankel linear system that the extended Euclidean algorithm solves in $O(n^2)$ operations, as a Padé-type problem. With noise, every exact solver returns the same solution $s + H_n(a)^{-1} e'$ of the square subsystem formed by the first n equations, where e' holds their errors; it is far from short, so a decoder must use the structure of the noise. Decoders that count errors, as for Reed–Solomon codes, cannot: about 62.5% of the coordinates of e are nonzero, while beyond the q-ary list-decoding capacity radius ρ, the solution of $H_q(\rho) = 1 - n/m$, which is 12.2% here, every code has exponentially many codewords near some words, whatever its structure; a random code of this shape has about 2^{4952} codewords within distance 480 of a given word. Decoders that use the small size of the errors are lattice attacks or algebraic attacks in the style of Arora–Ge. At toy sizes, XL solves Arora–Ge systems

from Hankel and from uniform matrices with the same number of samples, and their Macaulay matrices have the same rank instance by instance (Appendix E).

Low-rank Hankel methods. Hankel matrices are the working object of several classical methods: Prony's method and its descendants in signal processing and system identification, such as Kung's realization algorithm and the eigensystem realization algorithm (ERA) of modal analysis [25, 26], the Berlekamp–Massey algorithm for linear recurrences [29], and the spectral learning of weighted automata [10]. The last rests on a weighted analogue of the Myhill–Nerode theorem, due to Fliess and to Carlyle and Paz [15, 21]: the smallest weighted automaton realizing a series has as many states as its Hankel matrix has rank. All of these methods exploit a Hankel matrix of low rank, that is, a sequence with a short linear recurrence. Ours has none. For five public keys, $H_m(a)$ has full rank 640 modulo 2, and a mod 2 and b mod 2 have linear complexity 703 to 704 of 1,407 and 383 to 385 of 768, about half their lengths, as for random sequences. The real-valued methods, which truncate a singular value decomposition and keep the modes that pass confidence tests such as the modal amplitude coherence, fare no better. They must lift the residues to integers, which turns the equation into $b = H_m(a)\, s + e - qk$ with an unknown integer vector k. For 20 public keys, its entries have standard deviation 6.8 to 7.9 and carry about 4.9 bits each, 3,665 to 3,820 bits in all, against the 1,300 bits of the secret. Kung's algorithm finds no model order in b: the singular values of its Hankel matrix show no gap, and as many modes pass the coherence test as for a uniform sequence ($p > 0.15$), whereas five damped modes with 10% noise give a gap of 10.8, and all ten pass it. (The modal phase collinearity, the other usual test, needs several output channels.) The lift also hides what Berlekamp–Massey finds: a sequence from a linear recurrence of order 64 modulo q gives $H_m(a)$ of rank 64 modulo 2 but of full rank over the reals. The real singular values of $H_m(a)$ do differ from those of a uniform matrix, whose top ten hold 5.3 to 5.4% of the squared norm against 5.8 to 8.2%; this describes the integer lift of public data, not the problem modulo q, and the experiments of Appendix E, which measure the attack itself, show no difference. The same fact gives a design rule: a must come from a cryptographic XOF, never from a short linear recurrence. A sequence from a 64-cell shift register makes $H_m(a)$ of rank 63; then $H_m(a)\, s$ lies in a subspace of that dimension, and the dual attack works in a lattice of determinant q^{63} instead of q^{640}.

Power-of-two modulus. Reducing a sample modulo 2^t, for $t < 12$, gives an LWE instance with the same secret and error modulo 2^t. For small t such an instance does not even determine s. Modulo 2, the error is uniform, since odd values of CBD_2 have total probability 1/2, so b mod 2 is independent of s. Modulo 4 and 8, the error has entropy 1.906 and 2.031 bits, so each of the 768 samples carries at most 0.094 and 0.969 bits of information about s mod 2^t: at most 72 and 744 bits in total, against an entropy of 1,220 and 1,300 bits for s mod 4 and s mod 8. For larger t, the reduced instance has a smaller ratio of modulus to noise than the original, which makes lattice attacks harder, not easier: the lattice estimator puts the reductions modulo 16 and 32 at $2^{382.8}$ and $2^{316.5}$, against $2^{137.1}$ for the full instance. Algebraic attacks lose exactness in the ring as well: the error polynomial $\prod_{c=-2}^{2}(t - c)$, a product of five consecutive integers and so always divisible by 8, has 24 roots modulo 2^{12} rather than 5, among them 510, 1024 and 2046; Appendix E compares the resulting systems for Hankel and uniform matrices.

The ciphertext instance. In the ciphertext shape of Assumption 1, r meets two Hankel matrices, $H_{768}(a)$ and $H_{128}(b')$, where b' holds the first 767 entries of the (by then uniform) public key. Each block withholds $n - 1 = 639$ coefficients of its own convolution, so Proposition 2 and the functional analysis apply to each block. The second block embeds into $\mathbb{Z}_q[X]/(X^{767} - 1)$, with $767 = 13 \cdot 59$, whose irreducible factors have degrees 1, 12, 58, 348 and 348; its only low-degree factor, $X - 1$, is shared with the first block. Combining the blocks through it gives functionals (φ_1, φ_2) with $H_{768}(a)^\top \varphi_1 + H_{128}(b')^\top \varphi_2$ a multiple of $\mathbf{1}$: the lattice of the dual hybrid attack on

Attack family	Status	Reason	What an adaptation would need
Generic lattice attacks: primal, dual, hybrids, BKW, Arora–Ge	Estimated (Section 7)	Treated as unstructured LWE; the cheapest is the dual hybrid attack at $2^{137.1}$ (Core-SVP)	Not applicable: this is the baseline that a structural attack must beat
Any structure that lattice reduction exploits	Observed: none at reduced size	Success block sizes for $n = 100$ to 130 and reduced-basis profiles match those of uniform matrices (Appendix E)	A difference that appears only at larger dimensions
Output truncation [40]	Proved: does not apply	Every published output depends on all n secret coordinates, so discarding outputs removes no unknowns	Outputs that see only part of the secret, which the window excludes by construction
Low-rank Hankel methods: Prony, ERA, Berlekamp–Massey, spectral learning of automata	Observed: do not apply	They need a Hankel matrix of low rank; $H_m(a)$ has full rank, and a and b have the linear complexity of random sequences (Section 4.3)	A public sequence with a short linear recurrence, which SHAKE256 output does not have
Coefficient peeling [9]	Proved: does not apply	The low coefficients that involve few secret coordinates are withheld, and the outputs are noisy	Access to edge coefficients, or noiseless outputs
Evaluation at roots via $X^L - 1$ [20]	Estimated and observed: no gain	Needs the $n - 1$ withheld coefficients; exact functionals give lattices of the same determinant as the dual hybrid attack, with a costlier guess (Appendix C)	Functionals shorter than in uniform lattices of the same shape, or a cheaper guess
Shifted dual functionals (the analogue of ring rotations)	Estimated: at most about 1 bit	Shifted functionals satisfy shifted constraints, never in the same lattice; a free 640-fold supply of short vectors saves at most 1.0 bit (Section 4.3)	A use of shifted functionals within one lattice, or shifts that make vectors shorter
Modulus switching to 2^t	Proved for $2^t \leq 8$; estimated above	Modulo 2 the error is uniform; modulo 4 and 8 the samples carry too little information; larger t lowers the modulus-to-noise ratio ($2^{382.8}$ modulo 16)	A reduction that improves the ratio of modulus to noise
Subring BDD with the withheld coefficients as unknowns	Proved, for linear attacks: reduces to the window equations	Modulo any divisor of degree at most $n - 1$, the withheld coefficients mask the equation; eliminating them leaves $b = H_m(a)\, s + e$ (Proposition 2)	A non-linear use of the ring, such as norms or automorphisms, that avoids the withheld coefficients
Public key and ciphertexts jointly	Proved: reduces to Assumption 1	Hybrid argument over the ciphertexts; as independent samples, the decryption equations cost $2^{254.1}$ (coded-BKW) or $2^{301.6}$ (Core-SVP) (Section 5.3)	An attack on Assumption 1 itself
Biased functionals toward $\tilde{s} \bmod g$	Estimated and observed: no gain	Same determinant in k more dimensions, with the costlier guess; at reduced size Hankel and uniform lattices have the same shortest vectors (Appendix C)	A bias toward a non-linear function of s, or toward one cheaper to guess
Arora–Ge systems over $\mathbb{Z}_{16}$ and $\mathbb{Z}_{2^{12}}$	Observed: no difference at toy size	Exact solution counts for $n \leq 8$ agree for Hankel and uniform matrices; where repeated with 512 instances per kind, two one-sided tests bound the difference in the fraction of uniquely solvable instances below 0.054 (Appendix E)	A difference at larger dimensions, or for bit-level and SAT solvers
Noisy Padé decoding	Open; none observed at toy size	Counting errors cannot work: 62.5% are wrong, against a list-decoding capacity of 12.2%; using their size, XL behaves identically on Hankel and uniform systems at toy size (Section 4.3, Appendix E)	An algebraic decoder that exploits both the small errors and the Hankel structure

Table 2: Status of attack families against Windowed Hankel LWE at the parameters of Assumption 1

the stacked 896×640 matrix, with the costlier guess $r(1)$, 6.7 bits against 2.0 for one coordinate. That the public key and the first ciphertext component use the same matrix with different secrets is the usual situation of LWE with several secrets, and the proof of Theorem 1 makes b uniform before it touches u.

Reuse of the public sequence. The sequence a generates both the public key and the first component of every ciphertext, much as ML-KEM reuses its matrix. Each key pair has its own seed, so no structure is shared between users. Within one key, the public key and every ciphertext use the same matrix, $b = H_m(a)\, s + e$ and $u = H_m(a)\, r + e_1$, so a single lattice reduction, or a single set of short dual vectors, serves all of them. This does not lower the security of a key. Recovering s from the public key already breaks every ciphertext of that key, and attacking any one ciphertext costs at least as much, in the cost models of Section 7, as the public-key instance; reuse only spreads that one cost over the ciphertexts. The relevant figure is therefore the cost per key, as for ML-KEM, which shares its matrix in the same way.

4.4 Attack summary and what counts as a break

Table 2 collects the attack families considered in this paper. Each row says whether the family applies, why, and what an adaptation would need; the last row is the direction we consider open. The status says what kind of evidence supports it: *proved* for a mathematical argument about the attack as stated, *estimated* for cost models of the lattice estimator, *observed* for experiments at reduced size. None of these is a lower bound on every attack of the family.

What counts as a break. We distinguish two levels. Assumption 1 is refuted by any attack that uses the Hankel structure to solve search or decision Windowed Hankel LWE significantly faster than the best generic attack on LWE with a uniform matrix and the same parameters: for the public key, by more than a few bits below $2^{137.1}$ under the Core-SVP model, or by a margin that grows with n. Such an attack would matter even if it left the parameter set above its target. The parameter set itself is broken by an attack that costs less than the Category 1 reference, $2^{115.5}$ under the Core-SVP model or $2^{139.7}$ under MATZOV, the costs of the cheapest attack on Kyber-512 under the same models. Evidence at reduced dimensions also counts: an algorithm that solves Hankel instances measurably faster than uniform-matrix instances of the same size, as in the comparisons of Appendices C and E, would point to a structural weakness worth scaling up. To give such comparisons fixed targets, the repository includes challenge instances of Windowed Hankel LWE for $n = 64$ to 640, with $m = 1.2n$, $q = 4096$ and CBD_2; each has a uniform twin of the same shape and a hash commitment to its secret, and its matrix is derived from its name alone.

5 The Scheme

The parameters are $n = 640$, $W = 128$, $m = n + W = 768$, $q = 4096$ and $\eta = 2$; the public sequence has length $L = 2n + W - 1 = 1407$, and $H = H_m(a)$.

5.1 Public-key encryption

Algorithm 1 $\mathsf{CPA.KeyGen}(\sigma)$

Require: σ, 32 bytes

$(seed_a \,\|\, seed_e) \leftarrow \mathrm{SHAKE256}(\sigma \,\|\, \texttt{"_hankel_cpa_keygen_seed"}, 64)$
$a \leftarrow (\mathrm{SHAKE256}(seed_a, 2L)$ read as little-endian 16-bit words$) \bmod q$
$\beta \leftarrow \mathrm{SHAKE256}(seed_e \,\|\, \texttt{"_hankel_noise"}, n+m)$
$s \leftarrow \mathrm{CBD}_2(\beta[0{:}n])$; $e \leftarrow \mathrm{CBD}_2(\beta[n{:}n+m])$
if $s = 0$ **then** $s_0 \leftarrow 1$ ▷ without branching; probability $2^{-905.6}$
$b \leftarrow Hs + e \bmod q$
return $pk = (seed_a, b)$, $sk = s$

Algorithm 2 $\mathsf{CPA.Enc}(pk, M; c)$

Require: $pk = (seed_a, b)$, message $M \in \{0,1\}^{256}$, coins $c \in \{0,1\}^{256}$

Regenerate a from $seed_a$
$\gamma \leftarrow \mathrm{SHAKE256}(c \,\|\, \texttt{"_hankel_encaps_coins"}, n+m+W)$
$r \leftarrow \mathrm{CBD}_2(\gamma[0{:}n])$; $e_1 \leftarrow \mathrm{CBD}_2(\gamma[n{:}n+m])$; $e_2 \leftarrow \mathrm{CBD}_2(\gamma[n+m{:}n+m+W])$
$u \leftarrow Hr + e_1 \bmod q$
$K_w \leftarrow \sum_{j=0}^{n-1} r_j\, b_{w+j} \bmod q$ for $0 \le w < W$ ▷ $K = H_W(b)\, r$
$v \leftarrow K + \mathrm{PAM}(M) + e_2 \bmod q$
return (u, v)

Algorithm 3 $\mathsf{CPA.Dec}(s, (u, v))$

$K'_w \leftarrow \sum_{k=0}^{n-1} s_k\, u_{w+k} \bmod q$ for $0 \le w < W$ ▷ $K' = H_W(u)\, s$
return $\mathrm{Decode}(v - K' \bmod q)$ ▷ Proposition 1, coordinate-wise

Correctness. Substituting $b = Hs + e$ and $u = Hr + e_1$ gives

$$K_w = \sum_{j,k} a_{w+j+k}\, r_j s_k + \sum_j r_j e_{w+j}, \qquad K'_w = \sum_{j,k} a_{w+j+k}\, r_j s_k + \sum_k s_k e_{1,w+k}.$$

The double sums agree because each term depends only on $j+k$; this is the Hankel counterpart of the identity $r^\top(As) = (r^\top A)s$ behind LWE-based key exchange [27]. Hence

$$v_w - K'_w = \mathrm{PAM}(M)_w + \underbrace{\sum_{j=0}^{n-1} r_j e_{w+j} - \sum_{k=0}^{n-1} s_k e_{1,w+k} + e_{2,w}}_{e_{\mathrm{dec},w}} \pmod q,$$

and by Proposition 1 decryption succeeds whenever $-512 \le e_{\mathrm{dec},w} \le 511$ for every w. The term e_2 cannot be omitted: without it, $v - \mathrm{PAM}(M) = H_W(b)\, r$ would hand an attacker W noiseless linear equations in the ephemeral secret r.

5.2 The KEM

The KEM is the Fujisaki–Okamoto transform with implicit rejection $\mathsf{FO}^{\not\perp} = \mathsf{U}^{\not\perp} \circ \mathsf{T}$ of Hofheinz, Hövelmanns and Kiltz [23], applied to the encryption scheme, with its two random oracles instantiated by domain-separated SHAKE256:

$$G(M) = \mathrm{SHAKE256}(M \,\|\, \texttt{"_hankel_fo_G"}, 32),$$
$$H(X, ct) = \mathrm{SHAKE256}(X \,\|\, ct \,\|\, \texttt{"_hankel_fo_H"}, 32).$$

G gives the coins of the encryption and H the shared key. The secret key holds s as signed bytes, a rejection value z, and a copy of the public key.

Algorithm 4 KeyGen()

$\sigma \leftarrow$ 64 uniform bytes
$(pk, s) \leftarrow \mathsf{CPA.KeyGen}(\sigma[0{:}32])$
$z \leftarrow \mathrm{SHAKE256}(\sigma[32{:}64] \,\|\, \texttt{"_hankel_cca_reject"}, 32)$
return pk, $sk = (s, z, pk)$

Algorithm 5 Encaps(pk)

$M \leftarrow \{0,1\}^{256}$
$ct \leftarrow \mathsf{CPA.Enc}(pk, M; G(M))$
return ct, $K = H(M, ct)$

Algorithm 6 Decaps(sk, ct)

Parse $sk = (s, z, pk)$
$M' \leftarrow \mathsf{CPA.Dec}(s, ct)$
$ct' \leftarrow \mathsf{CPA.Enc}(pk, M'; G(M'))$
$X \leftarrow M'$ if $ct = ct'$, else z ▷ constant-time comparison and selection
return $K = H(X, ct)$

A ciphertext with the wrong length or a malformed header can never equal a re-encryption, so it is rejected implicitly like any other invalid ciphertext. Appendix B gives the byte-level formats.

5.3 Security of the scheme

We model SHAKE256 as a random oracle, so that its domain-separated uses, such as G and H, are independent random oracles, and measure an IND-CPA adversary by $\mathrm{Adv}^{\text{IND-CPA}}(\mathcal{A}) = |\Pr[\theta' = \theta] - 1/2|$, where θ is the challenge bit and θ' the adversary's guess, and a decision distinguisher by the difference of its acceptance probabilities on the two distributions. The problem of Definition 2 samples the public sequence uniformly, whereas the scheme expands it from a public seed; the random-oracle model is what connects the two.

Theorem 1 (IND-CPA)**.** For every IND-CPA adversary $\mathcal{A}$ against the encryption scheme of Algorithms 1 and 2 that makes at most Q random-oracle queries, there are distinguishers $\mathcal{B}_1$ for decision $\mathsf{WH\text{-}LWE}_{640,(768),q,\mathrm{CBD}_2,\mathrm{CBD}_2}$ and $\mathcal{B}_2$ for decision $\mathsf{WH\text{-}LWE}_{640,(768,128),q,\mathrm{CBD}_2,\mathrm{CBD}_2}$, running in about the time of $\mathcal{A}$, such that

$$\mathrm{Adv}^{\text{IND-CPA}}(\mathcal{A}) \le \mathrm{Adv}(\mathcal{B}_1) + \mathrm{Adv}(\mathcal{B}_2) + 3Q/2^{256} + 2^{-905.6}.$$

In the quantum random-oracle model, where $\mathcal{A}$ is a quantum algorithm that makes at most Q superposition queries to the oracle at query depth at most d, the same holds with quantum distinguishers $\mathcal{B}_1, \mathcal{B}_2$ and with $3Q/2^{256}$ replaced by $4\sqrt{3Q(d+1)}/2^{128}$.

Proof. Game 0 is the IND-CPA game. Game 1 replaces $(seed_a, seed_e)$, the noise derived from $seed_e$, and the triple (r, e_1, e_2) derived from the challenge coins by independent samples from the same distributions, and omits the fix-up of an all-zero secret. The view of $\mathcal{A}$ changes only if it queries the oracle at one of the three secret 256-bit inputs, the key-generation seed σ, $seed_e$ or the coins, or if $s = 0$, so the two games differ by at most $3Q/2^{256} + 2^{-905.6}$. Game 2 samples a uniformly and programs the oracle at $seed_a$ to return 16-bit words whose residues modulo q

are a, with uniform high bits. Since q divides 2^{16}, residues of uniform 16-bit words are uniform, and $seed_a$ is uniform and unknown to $\mathcal{A}$ until it receives the public key, so Games 1 and 2 are identical. Game 3 replaces b by a uniform vector; $\mathcal{B}_1$ receives (a, b), programs the oracle as in Game 2 and runs $\mathcal{A}$, so the difference is $\mathrm{Adv}(\mathcal{B}_1)$. In Game 3, the challenge consists of $u = H_m(a)\, r + e_1$ and $v - \mathrm{PAM}(M_\theta) = H_W(b)\, r + e_2$, where $H_W(b)$ uses only the first 767 entries of the uniform b: an instance of decision $\mathsf{WH\text{-}LWE}_{640,(768,128),q,\mathrm{CBD}_2,\mathrm{CBD}_2}$ with secret r. Game 4 replaces $(u, v - \mathrm{PAM}(M_\theta))$ by uniform vectors, a difference of $\mathrm{Adv}(\mathcal{B}_2)$, where $\mathcal{B}_2$ samples the last entry of b itself. In Game 4 the challenge is independent of θ.

In the quantum random-oracle model only two steps change. The first hop is an instance of the semi-classical one-way-to-hiding theorem [7, Theorem 1], with S the set of the three secret inputs σ, $seed_e$ and the coins, H the oracle of Game 0, and G the oracle that agrees with H outside S and takes fresh uniform values on S; the input z consists of the public key, the challenge bit θ and the ciphertext components u and $H_W(b)\, r + e_2$, all computed from the values of H on S, and the remaining step of the game, which adds $\mathrm{PAM}(M_\theta)$ once $\mathcal{A}$ has chosen its messages, is part of the oracle algorithm. Then $\mathcal{A}$ with oracle H plays Game 0 and with oracle G plays Game 1, and the two differ by at most $2\sqrt{(d+1)P_{\mathrm{find}}}$. The set S is independent of z and of G: the three inputs are uniform 256-bit strings, z depends on H only through its values on S, which G does not share, and $seed_e$ and $seed_a$ are independent halves of the uniform output at σ. By [7, Corollary 1], $P_{\mathrm{find}} \le 4Q \cdot 3/2^{256}$, which gives the term $4\sqrt{3Q(d+1)}/2^{128}$; the fix-up event $s = 0$ is classical and contributes $2^{-905.6}$ as before. The oracle is programmed at $seed_a$ once, before $\mathcal{A}$ runs, with a uniformly distributed value, so Games 1 and 2 remain identical under superposition queries; no adaptive reprogramming is involved. Finally, $\mathcal{B}_1$ and $\mathcal{B}_2$ simulate the oracle for $\mathcal{A}$ perfectly with a $2Q$-wise independent function [44], so they run in about the time of $\mathcal{A}$. □

The order of the steps matters. In a real ciphertext, the matrix $H_W(b)$ of v is generated by $b = Hs + e$ and is therefore related to the matrix H of u; an attack that exploited this relation to eliminate r would distinguish b from uniform, and so would break the public-key instance.

Lemma 1 (Injectivity)**.** Fix the coins as in the Fujisaki–Okamoto transform, $c = G(M)$, and model G and the expansion of coins into (r, e_1, e_2) by SHAKE256 as random oracles, with a uniform as in Theorem 1. Then, over the choice of a and of the oracles, encryption fails to be injective on messages with probability $\epsilon \le 2^{-7,169}$.

Proof. Let messages $M_0 \neq M_1$ give the ciphertexts $(u^{(0)}, v^{(0)})$ and $(u^{(1)}, v^{(1)})$, from the triples $(r^{(0)}, e_1^{(0)}, e_2^{(0)})$ and $(r^{(1)}, e_1^{(1)}, e_2^{(1)})$; superscripts index the two messages. If $r^{(0)} = r^{(1)}$, which includes the case $G(M_0) = G(M_1)$ of equal coins, then equal u forces $e_1^{(0)} = e_1^{(1)}$, and the vector $v^{(0)} - v^{(1)} = \mathrm{PAM}(M_0) - \mathrm{PAM}(M_1) + e_2^{(0)} - e_2^{(1)} \in \mathbb{Z}_q^W$ is nonzero: at a coordinate where the messages differ, the first difference is a nonzero multiple of 1024 modulo q, and the second has absolute value at most 4. Otherwise $r^{(0)} \neq r^{(1)}$, so the coins differ, $G(M_0) \neq G(M_1)$; in the random-oracle model the two triples are then independent CBD_2 samples, independent of a, and we bound the probability that $u^{(0)} = u^{(1)}$ jointly with $d = r^{(0)} - r^{(1)} \neq 0$. Write $d = 2^t d'$ with d' having an odd entry, so $t \le 2$. Then $H_m(a)\, d' = T_m(d')\, a$. Let d'_k be the first odd entry of d', and take the m columns $k, \ldots, k+m-1$ of $T_m(d')$: the entry in row i and column $k+j$ is d'_{k+j-i}, which is even below the diagonal ($j < i$, index below k) and odd on it. This $m \times m$ matrix is therefore upper triangular modulo 2 with odd diagonal, hence invertible modulo q, so $a \mapsto H_m(a)\, d'$ is onto $\mathbb{Z}_q^m$. Hence $H_m(a)\, d$ is uniform on $2^t\mathbb{Z}_q^m$ for uniform a, and equals $e_1^{(1)} - e_1^{(0)}$ with probability at most $(4/q)^{768} = 2^{-7,680}$. A union bound over fewer than 2^{511} pairs of messages gives the claim. □

We do not claim that encryption is γ-spread in the sense of [23] with large γ. That notion takes the probability over the encryption coins for a fixed oracle, and our coins are 256-bit strings,

so a fixed message has at most 2^{256} ciphertexts and $\gamma \le 256$. The 768 error coefficients of u carry far more entropy only when they are modeled as independent samples, which is a different probability space. Theorem 2 does not need spreadness.

Theorem 2 (IND-CCA, random-oracle model)**.** Let KEM be the scheme of Algorithms 4 to 6: the transform $\mathsf{FO}^{\not\perp}$ of [23] applied to the encryption scheme, with message space $\{0,1\}^{256}$ and $\delta \le 2^{-134.3}$ (Theorem 3). For every IND-CCA adversary $\mathcal{B}$ making at most q_G queries to G, q_H queries to H, Q other queries to SHAKE256 and any number of decapsulation queries, there is an IND-CPA adversary $\mathcal{A}$ against the encryption scheme, running in about the time of $\mathcal{B}$, such that

$$\mathrm{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{B}) \le 3\,\mathrm{Adv}^{\text{IND-CPA}}(\mathcal{A}) + (q_G + q_H)\,\delta + \frac{2q_G + 3q_H + Q + 1}{2^{256}}.$$

Proof. Theorem 3.4 of [23] bounds the IND-CCA advantage by $q_H/2^{256}$ plus the OW-PCA advantage of an adversary against T that makes at most q_H plaintext-checking queries. Theorem 3.2 of [23], in its corrected form and with no validity queries, bounds the latter by $(q_G + q_H)\,\delta + (2q_G + 1)/2^{256} + 3\,\mathrm{Adv}^{\text{IND-CPA}}$, with one adjustment. Its proof bounds by $q_G/2^{256}$ the probability that G is queried on the second challenge message, but in the game it simulates, the plaintext-checking oracle also queries G, as the corrected bound of their Theorem 3.1 accounts for; counting those queries as well replaces $(2q_G + 1)/2^{256}$ by $(2(q_G + q_H) + 1)/2^{256}$, a difference without numerical consequence. Without validity queries this is an OW-PCA bound, which, as noted before that theorem in [23], does not need the spread condition. Their δ is the expectation over key pairs of the worst case over messages of the failure probability over the encryption coins. For a fixed key pair that probability is the same for every message, because the decryption error e_{dec} does not involve M; the worst case therefore equals the average, and the expectation over key pairs is the probability over keys and coins that Theorem 3 bounds. Finally, $\mathsf{U}^{\not\perp}$ draws its rejection secret uniformly from the message space, whereas Algorithm 4 derives z from 32 secret uniform bytes through SHAKE256: z is uniform and independent of the adversary's view unless one of its Q other queries is that input, which adds at most $Q/2^{256}$. □

Combined with Theorem 1, the IND-CCA advantage of the KEM is bounded by three times the advantages against the two instances of Assumption 1, plus $(q_G + q_H)\,\delta$, which stays below 2^{-70} for 2^{64} queries, plus terms of order $Q/2^{256}$.

Remark 1 (Earlier versions)**.** Theorem 2 covers the implementation as specified here, from version 0.3 of the code. Versions up to 0.2 used a Kyber-style variant, which derived the coins and a pre-key from $\mathrm{SHAKE256}(M \,\|\, \mathrm{SHA\text{-}256}(pk), 64)$ and the key from the pre-key and $\mathrm{SHA\text{-}256}(ct)$, and which the theorem did not cover. Following the template exactly costs one SHAKE256 pass over the ciphertext in encapsulation and decapsulation (Section 8).

Remark 2 (Quantum random-oracle model)**.** By its quantum part, Theorem 1 bounds the IND-CPA advantage of quantum adversaries with superposition access to the oracle by the advantages of quantum distinguishers against the two instances of Assumption 1. Applying the quantum random-oracle results of Bindel et al. [11] (their Theorems 1 and 2 and Lemma 6) to the transform, we obtain, for an adversary making q quantum random-oracle queries at depth d and q_D classical decapsulation queries,

$$\mathrm{Adv}^{\text{IND-CCA}} \le 2\sqrt{(d+2)\left(\mathrm{Adv}^{\text{IND-CPA}}_{\pm} + \tfrac{8(q+1)}{2^{256}}\right)} + \left((4d+1)\,\delta + \sqrt{3\epsilon}\right)(q + q_D) + 2\epsilon + 4\sqrt{dq/2^{256}},$$

where $\mathrm{Adv}^{\text{IND-CPA}}_{\pm} = 2\,|\Pr[\theta' = \theta] - 1/2|$ is the quantum IND-CPA advantage in their normalization, at most twice the bound of Theorem 1 in the quantum random-oracle model, $\epsilon \le 2^{-7{,}169}$ is the injectivity bound of Lemma 1, and the last term covers implicit rejection with a hashed secret; q and d also bound the queries of the reductions, which run in about the same resources. The failure term $(4d+1)(q+q_D)\,\delta$ is about 2^{-20} when $d(q+q_D) = 2^{112}$.

Many ciphertexts. The argument extends to an attacker who sees the public key and any number Q of ciphertexts, by the usual hybrid argument. Once b is uniform, the ciphertexts are instances of decision $\mathsf{WH\text{-}LWE}_{640,(768,128),q,\mathrm{CBD}_2,\mathrm{CBD}_2}$ that share (a, b) and have independent secrets $r_1, \ldots, r_Q$, and a reduction can generate all but one of them itself. The advantage of any attack on the joint distribution is therefore at most its advantage against the public-key instance plus Q times its advantage against the ciphertext instance: attacks that combine a key with its ciphertexts are attacks on Assumption 1. The relation that decryption relies on illustrates this. Since $\mathrm{PAM}(M)$ is a multiple of $q/4$, each ciphertext gives $W = 128$ samples

$$v - H_W(u)\, s \equiv N \pmod{q/4}, \qquad N = e_2 + H_W(e)\, r - H_W(e_1)\, s,$$

of LWE in s with noise of variance $2n + 1$ on average over key generation, and an attacker can collect as many as it likes. For a fixed key the variance of coordinate w over the ciphertext randomness is $1 + \|s\|^2 + \sum_{j<n} e_{w+j}^2$, whose expectation is $2n + 1$. With $q/4 = 1{,}024$ against a standard deviation of 35.8, the cheapest lattice attack the estimator finds is the dual hybrid attack, at $2^{301.6}$ in the Core-SVP model; coded-BKW costs $2^{254.1}$ but needs about 2^{242} samples, or 2^{235} ciphertexts. The estimator treats these samples as independent, which they are not: their noise depends on s and e. The hybrid argument covers that dependence; the estimate only shows that the obvious use of many ciphertexts is far more expensive than attacking the public key.

6 Parameter Selection

Table 3 lists the parameters and sizes. We explain each choice below and compare it with nearby alternatives. We carried out this comparison after fixing the parameters; it confirms the choice, and it turned up one alternative that deserves further study.

Parameter	Value
Dimension n	640
Window W / published coordinates $m = n + W$	128 / 768
Modulus q	$4096 = 2^{12}$
Secret and error distribution	CBD_2 (variance 1)
Public sequence length $L = 2n + W - 1$	1,407
Message	256 bits (4-PAM over W coordinates)
Public key	1,196 B
Ciphertext	1,356 B
Secret key	1,868 B

Table 3: Parameters of Windowed Hankel KEM. The public key consists of a 12-byte header, a 32-byte seed and 768 coefficients of 12 bits; the ciphertext of a 12-byte header and $768 + 128$ coefficients of 12 bits.

Modulus. A power-of-two modulus makes reduction a bitwise mask and lets uniform coefficients be read directly from masked 16-bit words of SHAKE256 output, without rejection sampling. With $q = 2^{12}$, two coefficients also pack exactly into three bytes, so every byte string of the right length decodes to valid coefficients, and products can be accumulated in 16-bit integers (Section 8). The cost is that NTT-based multiplication modulo q itself is unavailable, although Karatsuba or Toom–Cook multiplication, or NTTs over an auxiliary modulus as used for NTT-unfriendly rings [18], remain possible.

Encoding and window. Carrying two bits per coordinate halves the window needed for a 256-bit message, from 256 coordinates to 128. Each window coordinate adds one coefficient to

each of b, u and v, so the smaller window saves 3×128 coefficients, or 576 bytes. In return, the decision threshold falls from $q/4$ to $q/8$, and the modulus must leave room for it.

Noise and modulus. Table 4 applies the Chernoff bound of Section 7.3 to several combinations, all with $n = 640$. With 4-PAM and CBD_2 noise, $q = 2^{12}$ is the smallest power of two that keeps the bound below 2^{-128}; at $q = 2^{11}$ the bound is only $2^{-28.5}$. Wider noise, CBD_3, would need $q = 2^{13}$ and one more bit per coefficient.

q	noise	bits per coordinate	threshold / σ	$\log_2$ DFR bound
2^{11}	CBD_1	2	14.30	−138.2
2^{11}	CBD_2	2	7.15	−28.7
2^{11}	CBD_2	1	14.31	−133.3
2^{12}	CBD_2	2	14.31	−134.3
2^{12}	CBD_3	2	9.54	−56.3
2^{13}	CBD_3	2	19.08	−234.8

Table 4: Chernoff bound on the decryption failure rate for $n = 640$ and a 256-bit message. The row with $q = 2^{12}$ and CBD_2 is the chosen parameter set.

Dimension. Table 5 shows how the cheapest primal and dual estimates move with n, and how the viable alternatives of Table 4 compare. At $n = 576$, the dual hybrid estimate would already be $2^{121.2}$ in the Core-SVP model, above Kyber-512's $2^{115.5}$; we take $n = 640$ to leave a margin for attacks that might exploit the structure.

			Core-SVP		MATZOV		
n	q	noise	primal	dual hybrid	primal	dual hybrid	PK + CT
512	2^{12}	CBD_2	108.3	105.4	134.1	129.6	2,168 B
576	2^{12}	CBD_2	125.3	121.2	150.3	145.1	2,360 B
640	$\mathbf{2^{12}}$	$\mathbf{CBD_2}$	**142.5**	**137.1**	**166.8**	**160.3**	**2,552 B**
704	2^{12}	CBD_2	159.7	153.3	183.4	176.1	2,744 B
640	2^{11}	CBD_1	141.6	134.3	165.9	157.7	2,344 B
640	2^{13}	CBD_3	136.9	133.7	161.7	157.4	2,760 B

Table 5: Estimated costs ($\log_2$) of the primal (uSVP) and dual hybrid attacks, and the combined size of public key and ciphertext including headers, assuming 11- and 13-bit coefficients pack as tightly as 12-bit ones. The chosen parameter set is in bold.

An alternative. The set $(n, q, \eta) = (640, 2^{11}, 1)$ is 208 bytes (8%) smaller and has a slightly lower failure bound, $2^{-137.8}$, for a dual hybrid estimate 2.8 bits lower. Its secret, though, is ternary with $\Pr[s_j = 0] = 1/2$, the regime in which combinatorial and hybrid attacks on small secrets improve, and the estimator's meet-in-the-middle optimizer did not converge for it in every cost model. We keep CBD_2 and leave this alternative to future work.

Higher categories. The same method yields exploratory parameters for NIST Categories 3 and 5, with $n = 936$ and $n = 1{,}208$, at the cost of one more bit per coefficient; Appendix D gives the derivation, and the C implementation covers both sets.

Comparison. Table 6 compares wire sizes with other LWE-based KEMs at the 128-bit level; for a like-for-like comparison, our figures exclude the 12-byte format headers. Ciphertexts are sent uncompressed. Rounding u and v to fewer bits, as ML-KEM does, would shrink them further, but would need a new failure analysis.

Scheme	Structure	Public key	Ciphertext	Total
ML-KEM-512 [32]	module-LWE	800 B	768 B	1,568 B
Windowed Hankel KEM (this work)	middle product	1,184 B	1,344 B	2,528 B
Titanium-CCA-Std128 [39]	MP-LWE	16,352 B	3,552 B	19,904 B
FrodoKEM-640 [13]	unstructured LWE	9,616 B	9,720 B	19,336 B

Table 6: Wire sizes of LWE-based KEMs targeting the 128-bit security level.

7 Security Analysis

7.1 Methodology

We treat each instance of Section 4 as unstructured LWE with secret dimension $n = 640$, modulus $q = 4096$, and secret and error of standard deviation 1, which amounts to assuming that the structure gives the attacker no advantage (Section 4.3 examines that assumption). Costs come from the lattice estimator [4] (commit `53da598`, August 2026) under two models for BKZ with block size β: the Core-SVP model [6], which charges $2^{0.292\beta}$ classically per call to an SVP oracle, and the more detailed model of MATZOV [30]. The same version of the estimator reproduces exactly the Kyber-512 figures in its documentation, and we computed every figure in this section in two SageMath installations (10.8 and 10.9) with identical results.

7.2 Generic attacks

Table 7 gives the estimates for the public key, which has $m = 768$ samples, alongside those for Kyber-512.

	Windowed Hankel KEM				Kyber-512	
	Core-SVP		MATZOV		Core-SVP	MATZOV
Attack	β	cost	β	cost	cost	cost
Primal (uSVP)	488	$2^{142.5}$	488	$2^{166.8}$	$2^{118.6}$	$2^{143.8}$
Primal (BDD)	488	$2^{143.2}$	471	$2^{163.1}$	$2^{119.4}$	$2^{140.2}$
Dual	507	$2^{148.0}$	509	$2^{173.7}$	$2^{123.8}$	$2^{149.9}$
Dual hybrid	468	$2^{137.1}$	459	$2^{160.3}$	$2^{115.5}$	$2^{139.7}$
Coded-BKW	—	—	—	$2^{207.3}$	—	$2^{178.8}$

Table 7: Lattice-estimator results for the public key ($n = 640$, $m = 768$, $q = 4096$, CBD_2 secret and error) and for Kyber-512, both treated as unstructured LWE. Costs are the estimator's total cost, including guessing in the hybrid attack. Coded-BKW is costed without lattice reduction, so it appears once per scheme.

The dual hybrid attack is the cheapest in both models. In the Core-SVP model it runs BKZ with $\beta = 468$, so lattice reduction alone costs $2^{0.292 \cdot 468} = 2^{136.7}$ classically and $2^{0.265 \cdot 468} = 2^{124.0}$ quantumly. The primal attack needs $\beta = 488$ in a lattice of dimension 1,194, at $2^{142.5}$ classically and $2^{129.3}$ quantumly; for Kyber-512, whose Category 1 claim rests on a classical Core-SVP cost of 2^{118} [14], the estimator gives a matching primal figure. These figures are generic-attack baselines, conditional on Assumption 1, not established security levels. Across the table, our instance costs at least 20 bits more than Kyber-512 under the same model, the smallest margin being 20.6 bits for the dual hybrid attack under MATZOV. This margin is the whole budget for an attack that exploits the structure; it is a comparison under a common cost model, not evidence of stronger practical security than ML-KEM. The most natural structural attack, through functionals derived from the cyclic embedding, is estimated to offer no gain over the dual hybrid attack because its guess is more expensive (Section 4.3 and Appendix C).

None of the remaining attack families comes close to the dual hybrid attack. The bounded support of the error lets each sample be written as a polynomial equation of degree 5 in s, which is the starting point of Arora–Ge attacks. Linearizing these equations would take about $\binom{n+5}{5} \approx 2^{39.7}$ samples, far more than the 768 available, and with 768 samples the estimator's Gröbner-basis model reaches a degree of regularity of 442, at a cost of about 2^{2100}. The primal hybrid attack is no cheaper than the plain primal attacks ($2^{143.2}$ and $2^{163.1}$ in the two models), and its meet-in-the-middle variant costs $2^{283.4}$. The meet-in-the-middle dual hybrid, designed for sparse or ternary secrets, costs $2^{748.0}$ against CBD_2.

Finally, none of the optimal attacks uses all the samples on offer: the primal attack uses 553, the dual attack about 620, and the dual hybrid attack 640, out of 768. The 896 samples that a ciphertext provides in the secret r therefore do not lower the estimates either.

Fast products. Products with a Hankel matrix cost $O((m+n)\log(m+n))$ operations by FFT instead of mn, so we checked where an attacker could use them. Lattice reduction does not multiply by A; the one way the structure could help it, shifts of short vectors, is bounded in Section 4.3 at about one bit. In the dual hybrid attack, the work per guessed candidate, an FFT over t coordinates and table updates over the N short dual vectors, does not involve A either. Products with A occur once, when the dual vectors are multiplied by the $\zeta + t$ guessed columns: about $Nm(\zeta+t) = 2^{111.4}$ operations at the Core-SVP optimum ($N = 2^{96.4}$, $\zeta + t = 50$) and $2^{110.1}$ in the MATZOV model. The estimator does not charge this step at all, and it lies more than 25 bits below the totals, so fast products cannot lower the estimates.

7.3 Decryption failure rate

Theorem 3. The decryption failure rate satisfies $\delta \le 2^{-134.3}$.

Proof. For each window coordinate w, $e_{\mathrm{dec},w}$ is a sum of 1,281 independent terms: 1,280 products XY of independent CBD_2 variables, namely $r_j e_{w+j}$ and $-s_k e_{1,w+k}$ (the distribution of XY is symmetric), and the CBD_2 variable $e_{2,w}$. Each term has mean 0 and variance 1, so $\sigma = \sqrt{1281} \approx 35.79$ and the threshold 512 lies 14.31σ from the mean. The Chernoff bound with the exact moment-generating functions (Appendix A) gives $\Pr[e_{\mathrm{dec},w} \ge 512] \le 2^{-142.07}$ and, since the decoding interval is asymmetric and $e_{\mathrm{dec},w}$ is symmetric, $\Pr[e_{\mathrm{dec},w} \le -513] = \Pr[e_{\mathrm{dec},w} \ge 513] \le 2^{-142.61}$. So $e_{\mathrm{dec},w}$ leaves $[-512, 511]$ with probability at most $2^{-141.31}$, and a union bound over the $W = 2^7$ coordinates gives $\delta \le 2^{-134.31}$. The probability is over key generation and the encryption randomness. □

As usual, the bound is an average over key pairs. FIPS 203 lists $2^{-138.8}$ for ML-KEM-512 [32, Table 1]. Experiments cannot confirm a rate this small, but they agree with the bound: over 500 end-to-end exchanges (64,000 decoded coordinates), the largest $|e_{\mathrm{dec}}|$ was 131, and its mean and standard deviation, 28.4 and 21.4, match the values 28.5 and 21.6 predicted by a normal approximation.

Failure rate matters beyond correctness, because attacks that collect decryption failures can erode the security of LWE-based schemes whose failure rate is relatively high [19]. The 128 window coordinates share r, s and e_1, so their errors are correlated. The union bound in Theorem 3 remains valid, and is if anything conservative, but a failure, should one occur, reveals more about the secret than an independent-coordinate model would suggest. At $\delta \le 2^{-134.3}$ this is moot in practice: an attacker who submits 2^{64} honestly generated ciphertexts expects fewer than 2^{-70} failures.

8 Implementation

We implemented the KEM twice: a Python reference and a C implementation of the complete scheme. The C code is bit-exact with the reference. It reproduces all fifteen known-answer test

vectors, which come from a custom SHAKE-based generator rather than the NIST DRBG, and matches the reference on random inputs, on every branch of its seed handling and on malformed ciphertexts.

8.1 Arithmetic

All three Hankel products have the form $y_i = \sum_{j<n} a_{i+j}x_j \ (+\, e_i) \bmod q$. Because $q = 2^{12}$ divides 2^{16}, they can be computed modulo 2^{16}, using the natural wraparound of 16-bit integers, and reduced modulo q by a single mask at the end; this is exact for any 16-bit inputs, including negative secret coefficients in two's complement. On ARMv8, the kernel computes 64 outputs at a time in eight 8-lane accumulators. For each secret coefficient x_j it adds $x_j \cdot (a_{i+j}, \ldots, a_{i+j+63})$ to the accumulators with `vmlaq_laneq_u16`, so a 768-output product takes $768 \cdot 640/8 = 61{,}440$ multiply-accumulate instructions, and the secret only ever enters as a multiplication operand. A portable scalar kernel is kept and tested alongside.

8.2 Hashing

SHAKE256 uses a Keccak-$f[1600]$ permutation with fixed loop counts and no secret-dependent table lookups. It is the only hash function of the scheme: G, H and every expansion are domain-separated SHAKE256 calls.

8.3 Constant-time execution

No secret value selects a branch or a memory address. The all-zero-secret fix-up of Algorithm 1, the comparison $ct = ct'$ and the selection of X in Algorithm 6 use masks behind a compiler value barrier, so that the compiler cannot turn them back into branches. Every step that handles secrets is compiled as a separate function, which makes two kinds of check possible.

The first is a static audit of the compiled ARM64 code: a flow-sensitive dataflow analysis tracks which registers, stack slots and condition flags may depend on secret data, and reports any conditional branch on such a value, any division and any unexpected call. Across the nine secret-handling functions of the Category 1 code (1,167 instructions with Apple clang 21), it finds 30 conditional branches, all on public values, and no divisions or calls; the code of all three parameter sets of Appendix D, 25 functions and 2,970 instructions, passes the same audit. It does not check memory addressing, but every index in these functions is a loop counter, which we verified by inspection. Nor does it check that the hardware executes the instructions in data-independent time: like most constant-time code, ours assumes that the multiplication, multiply-accumulate and SIMD instructions it uses have data-independent latency, which ARMv8.4 makes explicit with its DIT mode and which we have not enabled.

The second is a set of fixed-versus-random Welch t-tests with about 4,000 calls per class, varying only the secret input: the seed for key generation, the message entropy for encapsulation, and, for decapsulation of a fixed ciphertext, the secret key, so that one class accepts and the other rejects. All inputs are prepared before timing, and all three tests give $|t| < 4.5$. A dedicated decapsulation test with 20,000 calls per class, repeated three times for each parameter set, gave $|t| \le 2.1$ between accepting and rejecting, $|t| \le 1.9$ for a control that splits one class in two, and medians at most three ticks of the 41 ns timer apart, out of 23 to 55 μs: decapsulation takes the same time whether it accepts or rejects. Like all such tests, they can only detect differences above their statistical resolution, which a 41 ns timer limits; they complement the static audit rather than replace it. This evidence covers timing only, on a single core with a single compiler; we have not studied power or electromagnetic leakage.

8.4 Performance

On an Apple M3 Max, with Apple clang 21 at `-O3`, the median latencies over 5,000 calls with fresh inputs are 14 μs for key generation, 20 μs for encapsulation and 23 μs for decapsulation, or 57 μs for a full handshake. An encapsulation spends about 8 μs in the two Hankel products and about 10.5 μs in SHAKE256: 3.6 μs expanding the public sequence, 2.1 μs expanding the coins, and 4.6 μs in H, which absorbs the whole ciphertext. The Kyber-style transform of earlier versions hashed the ciphertext with SHA-256 instructions instead, and took 52 μs per handshake with the same compiler (Remark 1). Faster Keccak would help, and so would subquadratic multiplication: our products are quadratic, but Karatsuba or Toom–Cook multiplication, or NTTs over an auxiliary modulus [18], would reduce their cost. For comparison, we timed ML-KEM-512 and FrodoKEM-640 from liboqs 0.16.0, built for this machine with its ARMv8 code paths and the AES and SHA-2 instructions, by the same method and in the same session: a handshake takes 19.5 μs for ML-KEM-512, 740 μs for FrodoKEM-640-AES and 3.3 ms for FrodoKEM-640-SHAKE, against 60 μs for Windowed Hankel KEM in that session. Our scheme is thus about three times slower than ML-KEM-512, which multiplies by NTT in optimized assembly, and 12 to 54 times faster than FrodoKEM-640.

9 Limitations and Open Questions

The central limitation is the one we accepted by design: Windowed Hankel LWE has no reduction, and its hardness is an assumption. The estimates of Section 7 hold only if the structure gives an attacker no advantage, and the roughly 20-bit margin over Kyber-512 is all that a structural attack would need to overcome. We have checked the published structure-exploiting attacks, the exact and biased functionals and the subrings that arise from the embedding into $\mathbb{Z}_q[X]/(X^L-1)$, shifted functionals, algebraic decoding, and lattice reduction at reduced dimensions, and found no advantage under the models and experiments examined, but this does not rule out other attacks. We would look first at the open direction of Table 2, an algebraic decoder that exploits both the small errors and the Hankel structure, beyond what XL sees at toy sizes, and at the non-linear uses of the ring structure that Proposition 2 does not cover, including functionals biased toward non-linear functions of the secret. We also have no search-to-decision reduction, and the usual arguments favor a prime modulus. The setting has received little study, and the fate of Compact-LWE [12] counsels caution; we offer the scheme as a problem for cryptanalysis, not for use, until it has been analyzed independently.

Three questions seem central to us. Does random Hankel structure introduce dependencies that an attacker can exploit, at larger dimensions or with algorithms other than lattice reduction? Does withholding the edge coefficients block direct algebraic attacks in general, and not only those tested here, while keeping the efficiency of structured multiplication? And could a reduction be found for a public long operand, perhaps for other distributions or parameters?

Two further limitations are practical. Our implementation multiplies in $O(nm)$ operations, about 4.9×10^5 multiply-accumulates at $n = 640$. That is fast enough here; subquadratic methods would matter more at higher security levels. Our constant-time evidence covers only timing, on one platform (Section 8).

10 Conclusion

Windowed Hankel LWE publishes a single long sequence and keeps the secret short. This gives up the MP-LWE reductions, and in exchange an LWE-based KEM needs only 2.5 KB per exchange, without reducing products modulo any polynomial. If the structure gives an attacker nothing, generic attacks cost at least 20 bits more than against Kyber-512; this is a conditional baseline, not an established security level. The published structural attacks do not apply as published.

For the most natural new one, through the hidden cyclic embedding, and for its subrings, biased and shifted functionals, algebraic decoding and experiments at reduced dimensions, we found no advantage under the models and experiments examined. The structure has yet to face sustained independent cryptanalysis. We hope that the precise statement of the problem, the attack summary of Table 2, the criteria of Section 4.4 and the challenge instances make that analysis easier to undertake.

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A The Chernoff Bound

Let $Y = XX'$ for independent $X, X' \sim \mathrm{CBD}_2$. Its distribution is

$$\Pr[Y=0] = \tfrac{156}{256}, \quad \Pr[Y=\pm1] = \tfrac{32}{256}, \quad \Pr[Y=\pm2] = \tfrac{16}{256}, \quad \Pr[Y=\pm4] = \tfrac{2}{256},$$

with mean 0 and variance 1. Write $e_{\mathrm{dec},w} = \sum_{i=1}^{1280} Y_i + Z$ with $Z \sim \mathrm{CBD}_2$. Markov's inequality applied to $e^{te_{\mathrm{dec},w}}$ gives, for every $t > 0$,

$$\Pr[e_{\mathrm{dec},w} \geq 512] \leq \exp\bigl(1280 \ln M_Y(t) + \ln M_Z(t) - 512\, t\bigr),$$

where $M_Y(t) = \mathbb{E}[e^{tY}]$ is computed exactly from the distribution above and $M_Z(t) = \cosh^4(t/2)$. The exponent is convex in t and minimized at $t^* \approx 0.3715$, where the bound equals $2^{-142.07}$. The decoding interval $[-512, 511]$ is not symmetric, so the lower tail is bounded separately, by the same computation with threshold 513: $2^{-142.61}$. The same computation, with 1280 replaced by $2n$, the thresholds by $T = q/2^{b+1}$ and $T+1$ for b bits per coordinate and the union bound taken over $256/b$ coordinates, produces every entry of Table 4 and the failure rates of Appendix D.

B Wire Formats and Derivation Tags

All multi-byte integers are little-endian, and coefficients are packed in pairs (x_0, x_1) into the three bytes $(x_0 \bmod 2^8,\ \lfloor x_0/2^8 \rfloor + 16(x_1 \bmod 2^4),\ \lfloor x_1/2^4 \rfloor)$. Table 8 gives the layouts.

Object	Bytes	Layout
Public key	1,196	`"HNPK"`, then n, W, q, 0 as 16-bit integers (12 bytes); $seed_a$ (32); b packed (1,152)
Ciphertext	1,356	`"HNCT"`, then n, W, q, 0 (12 bytes); u packed (1,152); v packed (192)
Secret key	1,868	s as signed bytes (640); z (32); public key (1,196)

Table 8: Byte layouts.

Each SHAKE256 call is domain-separated by an ASCII tag appended to its input, listed in Table 9. The first six belong to the algorithms of Section 5. The last two exist only for deterministic testing, and the known-answer tests use them: the implementations accept a key-generation seed of any length, expanding one shorter than 64 bytes first, and can derive the encapsulated message from caller-supplied entropy.

Tag	Use
`_hankel_cpa_keygen_seed`	$(seed_a \| seed_e)$ from σ[0:32] (Algorithm 1)
`_hankel_noise`	s and e from $seed_e$ (Algorithm 1)
`_hankel_encaps_coins`	r, e_1 and e_2 from the coins c (Algorithm 2)
`_hankel_cca_reject`	rejection value z from σ[32:64] (Algorithm 4)
`_hankel_fo_G`	coins $G(M)$ (Algorithms 5 and 6)
`_hankel_fo_H`	shared key $H(X, ct)$, with $X = M$ or z (Algorithms 5 and 6)
`_hankel_cca_keygen_seed`	testing: expands a key-generation seed shorter than 64 bytes
`_hankel_encaps_entropy`	testing: $M = \mathrm{SHAKE256}(\rho \| \mathrm{tag}, 32)$ from entropy ρ

Table 9: Domain-separation tags.

C Short Functionals from the Cyclic Embedding

This appendix gives the computations behind the embedding argument of Section 4.3.

The lattice. Let g be a monic factor of $X^L - 1$ over $\mathbb{Z}_q$ of degree k, and let $R_g \in \mathbb{Z}_q^{k \times n}$ have as its column j the coefficient vector of $X^{n-1-j} \bmod g$, so that $R_g s$ is the coefficient vector of $\tilde{s} \bmod g$. The last k columns of R_g are unit vectors, so the condition $H_m(a)^\top \varphi = R_g^\top c$ determines c from k coordinates of $H_m(a)^\top \varphi$ and reduces to $n - k$ linear conditions $B_g \varphi \equiv 0 \pmod q$, with $B_g \in \mathbb{Z}_q^{(n-k) \times m}$. The functionals form the lattice $\Lambda_g = \{\varphi \in \mathbb{Z}^m : B_g \varphi \equiv 0 \pmod q\}$; for $k = 0$ it is the lattice of the plain dual attack. If B_g has full rank modulo 2, it maps $\mathbb{Z}^m$ onto $\mathbb{Z}_q^{n-k}$ and $\det \Lambda_g = q^{n-k}$, as for a uniform matrix, and as for the dual-hybrid lattice obtained by guessing k coordinates of s. We estimate the shortest vector by the Gaussian heuristic $\mathrm{GH}(\Lambda_g) = \Gamma(m/2+1)^{1/m} \det(\Lambda_g)^{1/m} / \sqrt{\pi}$.

The guess. Each factor g divides Φ_d for some $d \in \{1, 3, 7, 21\}$, and $X^i \bmod \Phi_d$ over $\mathbb{Z}$ is periodic in i with small coefficients. So $\tilde{s} \bmod g$ is a function of the integer vector $y = \tilde{s} \bmod \Phi_d$, whose coefficients are sums of about n/d entries of s. We estimate its entropy by the differential entropy $\frac{1}{2} \log_2 \det(2\pi e \, RR^\top)$ of a Gaussian with the covariance of y, where R is the integer matrix of $s \mapsto y$, capped at 12 bits per coefficient of $\tilde{s} \bmod g$. This bounds the entropy of $\tilde{s} \bmod g$ from above. For the cubic and sextic factors the bound is close: the reduction from y to $\tilde{s} \bmod g$ modulo q has a kernel lattice whose shortest vector has length about 78, against a standard deviation of 13.5 per coordinate of y, so distinct likely values of y almost never collide, and a Monte Carlo estimate of the collision entropy of $\tilde{s} \bmod g$ for the cubic factors gives 31.9 bits, a lower bound on its Shannon entropy, against the bound of 33.2. A previous version of this paper applied the Gaussian estimate to R_g itself, whose 2-adic entries are large, and obtained 36.0 bits, the cap, for the cubic factors and 71.0 bits for the sextic ones, which overstates the entropy of these factors.

Full size. Table 10 covers $n = 640$, $m = 768$ and five random sequences a. In every case B_g has full rank modulo 2, so the structured lattices have exactly the determinants of the corresponding dual-hybrid lattices, and under the Gaussian heuristic the comparison reduces to the guess. The Gaussian heuristic also places the shortest functionals above q: $\varphi^\top e$ then has standard deviation larger than the modulus, so a single functional reveals almost nothing, and any use must combine

many of them, as the dual attack does. We do not convert these figures into an attack cost; that is what the estimates of Section 7 provide for the dual hybrid attack, which, under the Gaussian heuristic, these functionals do not appear to beat.

		full rank			bits to guess	
Factor g	k	mod 2	$\log_2 \det$	GH	$\tilde{s}$ mod g	k coords of s
none (plain dual)	0	5/5	7,680	6,902	—	—
$X-1$	1	5/5	7,668	6,827	6.7	2.0
Φ_3	2	5/5	7,656	6,754	12.6	4.1
factors of Φ_7 (each)	3	5/5	7,644	6,681	33.2	6.1
factors of Φ_{21} (each)	6	5/5	7,608	6,467	61.7	12.2

Table 10: Lattices of functionals at full size ($n = 640$, $m = 768$, $q = 4096$, five random sequences). "Full rank mod 2" counts the sequences for which B_g has full rank modulo 2; GH is the Gaussian heuristic in the dual lattice's own units; the guess for $\tilde{s}$ mod g is the entropy bound described above, against 2.031 bits per coordinate of s. For composite divisors the same estimate gives 12.6 and 17.7 bits for Φ_3 and X^3-1, 33.2 and 37.1 for Φ_7 and X^7-1, and 61.7 and 94.7 for Φ_{21} and $X^{21}-1$, against 4.1, 6.1, 12.2, 14.2, 24.4 and 42.7 bits for as many coordinates of s.

Reduced sizes. To test whether Hankel structure produces unusually short functionals, we computed exact shortest vectors at reduced sizes, with fpylll enumeration after BKZ preprocessing, both for B_g derived from random Hankel matrices and for uniform matrices of the same shape. We chose $(n, W) = (40, 5)$, $(48, 10)$ and $(56, 15)$, giving lattice dimensions 45, 58 and 71, so that 21 divides L and all six factors above divide $X^L - 1$. Instances in which B_g was not of full rank modulo 2 were resampled; this happened once for a Hankel matrix and six times for a uniform one, all at the smallest size. Table 11 summarizes the ratio λ_1/GH over the seven targets of Table 10; the evidence directory gives the figures per target.

				Hankel		uniform	
n	W	m	instances per target	means	smallest	means	smallest
40	5	45	20	0.990–1.016	0.888	0.997–1.016	0.903
48	10	58	20	0.996–1.010	0.886	0.998–1.008	0.945
56	15	71	5	0.991–1.010	0.960	0.995–1.019	0.958

Table 11: Exact shortest vectors of the lattices of functionals, as a ratio to the Gaussian heuristic: range of the per-target means over the seven targets (including the plain dual lattice), and the smallest ratio over all instances of that size.

The per-target means agree with those of uniform matrices to within about 2%. The smallest Hankel ratio, 0.886, lies within the spread of the individual instances: at the smallest size, Hankel and uniform matrices reach 0.888 and 0.903. We found no sign that Hankel structure shortens the functionals. These comparisons are evidence about the attacks evaluated: similar determinants and shortest-vector statistics do not by themselves establish that the corresponding problems are equally hard.

Biased functionals. The biased functionals of Section 4.3 form the lattice $\{(\varphi, w) \in \mathbb{Z}^{m+n} : E_g(H_m(a)^\top \varphi - w) \equiv 0 \pmod q\}$, where $E_g \in \mathbb{Z}_q^{(n-k)\times n}$ has kernel $\{R_g^\top c\}$; built from the unit columns of R_g, it contains an identity block, so the determinant is always q^{n-k}. The lattice contains the trivial vectors $(0, R_g^\top c)$, for which the identity reads $0 = 0$. For $X-1$ and Φ_3 they are very short, since $X^i \bmod g$ has tiny coefficients, and a plain shortest-vector computation finds only them. We therefore project the lattice orthogonally to the k vectors $(0, R_g^\top e_t)$, with centered

integer entries, and measure the projection, of rank $m+n-k$, against its own Gaussian heuristic, computed from its covolume $q^{n-k}/\sqrt{\det(SS^\top)}$, where the rows of S are those k vectors. The projection leaves φ unchanged and replaces w by its distance to the real span of $R_g^\top$, which allows a non-integral c; a projected length is therefore a slight lower bound on the useful length, and both kinds of matrix are treated alike. Other trivial vectors $(0, w)$, with w congruent to some $R_g^\top c$ modulo q but outside the projected span, survive the projection, so we checked every vector found: all 840, over the three sizes, seven targets and both kinds of matrix, have $\varphi \neq 0$. The dimension is always $m+n = L+1$, and $21 \mid L$ leaves $L = 63$ or 84: we compute exact shortest vectors for $(n, W) = (24, 16)$, in dimension 64, and the first vector after BKZ-40 for $(32, 21)$ and $(40, 5)$, in dimension 85, with 20 instances of each kind per target. Over the seven targets and three sizes, the mean ratios for Hankel and uniform matrices differ by at most 0.015 for exact shortest vectors and 0.018 after BKZ-40, in both directions, and none of the 21 differences is significant (smallest $p = 0.086$, two-sided Mann–Whitney).

The scripts and their outputs are in the accompanying evidence directory.

D Higher Security Categories

This appendix derives exploratory candidate parameters for NIST Categories 3 and 5 by the method of Section 6. Larger dimensions raise the generic estimates but do not address the structural assumption. We size them as the Category 1 set is sized: the cheapest estimated attack should exceed that on the corresponding Kyber parameter set by at least 20 bits in both cost models, and the failure rate should be no larger than ML-KEM's. Under the estimator and models of Section 7, the cheapest attacks on Kyber-768 and Kyber-1024 cost $2^{174.3}$ and $2^{241.8}$ (Core-SVP) and $2^{196.4}$ and $2^{262.3}$ (MATZOV); FIPS 203 lists failure rates of $2^{-164.8}$ and $2^{-174.8}$ for ML-KEM-768 and ML-KEM-1024 [32, Table 1]. The message stays at 256 bits.

The Category 1 recipe does not scale. The decryption error has variance $2n+1$, so the failure rate grows with n. With $q = 2^{12}$, CBD_2 and 4-PAM, the Chernoff bound of Theorem 3 exceeds $2^{-164.8}$ for every $n \geq 640$; at $n = 640$ it is already $2^{-134.3}$. Four modifications reach the targets:

A. $q = 2^{13}$, keeping CBD_2 and 4-PAM, at the cost of one more bit per coefficient;

B. one bit per coordinate ($W = 256$), keeping $q = 2^{12}$ and CBD_2;

C. CBD_1 secret and noise, keeping $q = 2^{12}$ and 4-PAM;

D. $q = 2^{13}$ with CBD_3 and one bit per coordinate.

Table 12 gives the smallest n that meets the targets in each family. Family C is the smallest, but its secret is the sparse ternary one set aside in Section 6. Families A and B are within 102 bytes of each other at both levels. We choose family A because it keeps the Category 1 scheme unchanged except for q, and because both of its lengths L are prime with $X-1$ as the only small factor of $X^L - 1$; for family B at Category 5, $L = 2{,}511$ gives factors of degrees 2, 5, 6, 10 and 18.

	Category 3		Category 5	
Family	n	PK + CT	n	PK + CT
A: $q = 2^{13}$, CBD_2, 4-PAM	936	3,722 B	1,208	4,606 B
B: $q = 2^{12}$, CBD_2, 1 bit	872	3,824 B	1,128	4,592 B
C: $q = 2^{12}$, CBD_1, 4-PAM	960	3,512 B	1,280	4,472 B
D: $q = 2^{13}$, CBD_3, 1 bit	896	4,216 B	1,152	5,048 B

Table 12: Smallest n meeting the Category 3 and 5 targets, by family, with the combined size of public key and ciphertext including headers.

Proposed sets. Table 13 lists the family A sets. In both, the dual hybrid attack remains the cheapest, the margin over Kyber is at least 21 bits in both models, and the failure rate is far below the target, which leaves room for compressing ciphertexts later. Merely matching Kyber, without the 20-bit margin, would require $n = 848$ and $n = 1{,}120$, at 3,436 and 4,320 bytes.

	Category 3	Category 5
n, W, $m = n + W$	936, 128, 1,064	1,208, 128, 1,336
q, distribution, encoding	2^{13}, CBD_2, 4-PAM	2^{13}, CBD_2, 4-PAM
Dual hybrid, Core-SVP (β)	$2^{196.8}$ (674)	$2^{263.5}$ (902)
Dual hybrid, MATZOV (β)	$2^{218.2}$ (665)	$2^{283.7}$ (899)
Primal (uSVP), Core-SVP	$2^{205.9}$	$2^{277.7}$
Failure rate (Chernoff bound)	$2^{-370.4}$	$2^{-292.1}$
Public key / ciphertext	1,773 B / 1,949 B	2,215 B / 2,391 B
PK + CT without headers (ML-KEM)	3,698 B (2,272 B)	4,582 B (3,136 B)
$L = 2n + W - 1$	1,999 (prime)	2,543 (prime)

Table 13: Proposed parameters for Categories 3 and 5 (lattice estimator, commit `53da598`).

Ring structure. The dimension n of Windowed Hankel LWE can be any integer, and the factorization of $X^L - 1$ varies widely with it. For example, $n = 960$ in family A gives $L = 2{,}047 = 2^{11} - 1$, for which $X^L - 1$ has 186 irreducible factors of degree 11. The proposed sets happen to fall on the favorable side: $L = 1{,}999$ and $L = 2{,}543$ are prime, and 2 has order 333 and 1,271 modulo them, so $X - 1$ is the only factor of $X^L - 1$ of degree below 333. Of the low-degree factors examined in Appendix C, only $X - 1$ remains. Repeating that check at the new sizes gives the same result: for five random sequences each, the lattices of functionals for the plain dual attack and for $X - 1$ have full rank modulo 2, hence the determinants of the corresponding dual-hybrid lattices, while isolating $s(1)$ requires guessing 7.0 and 7.2 bits against 2.0 bits for a coordinate of s. Choosing L prime with 2 of large order is a free design rule for any further parameter set.

Implementation. Both sets use the C code of the Category 1 set (Section 8), compiled once per set. Since 2^{13} divides 2^{16}, 16-bit accumulation stays exact; only the packing, at 13 bits per coefficient and least significant bit first (which for 12 bits is the pair packing of Appendix B), and the 4-PAM scaling by $q/4$ change. To keep the sets apart, every derivation tag of Appendix B is followed by a domain string, `_wh936` or `_wh1208`; the tags of the Category 1 set are unchanged. Each set has 15 known-answer vectors, generated by a parameterized Python reference, and the C code reproduces them exactly. It passes the same static branch audit and fixed-versus-random timing tests as the Category 1 code, with $|t| < 4.5$ in all six tests. On the machine of Section 8, median latencies over 5,000 calls are 30, 31 and 37 μs for key generation, encapsulation and decapsulation at Category 3, and 40, 48 and 48 μs at Category 5, or about 100 and 135 μs for a full handshake.

What scaling does not change. These sets rest on the analogue of Assumption 1, for which a larger n adds no evidence. The criteria of Section 4.4 carry over, with Table 13 as the generic baseline and Kyber-768 and Kyber-1024 as references, and the open questions of Section 9 matter more, not less, at these levels.

E Experiments at Reduced Dimensions

This appendix applies the last criterion of Section 4.4. If lattice reduction or algebraic solving could exploit the Hankel structure, Hankel instances would fall at smaller block sizes, reduce to shorter vectors, or be solved from fewer samples than uniform instances of the same shape. We use $q = 4096$, CBD_2 secret and error, and BKZ 2.0 [16] in fpylll 0.6.4 with its default strategies.

Primal attack. For each n we draw instances $b = As + e$ in which A is either $H_m(a)$ or uniform, with the same m, and run progressive BKZ on the embedding lattice, two tours per block size, until a basis vector equals $\pm(-e, s, -1)$. We choose m to minimize the block size predicted by the 2016 estimate. Every instance was solved (Table 14). The differences are below one unit of block size, change sign with n and are not significant; the observed block sizes exceed the predictions equally for both kinds. A non-significant test does not by itself show equivalence, so we also bound the difference: the 95% Welch confidence intervals for the mean block size of Hankel minus uniform instances are $[-0.68, 1.26]$, $[-1.73, 0.81]$, $[-0.66, 2.54]$ and $[-1.55, 1.47]$ for $n = 100$ to 130, so an advantage of the Hankel structure larger than 1.8 block sizes is excluded at that level for every n, and a two one-sided tests procedure at the 5% level gives equivalence margins of 1.1 to 2.3.

				Success block size, mean (sd)			
n	m	d	predicted	Hankel	uniform	trials	p
100	87	188	10	19.83 (2.37)	19.54 (2.42)	48	0.61
110	119	230	15	26.19 (3.13)	26.65 (3.14)	48	0.52
120	124	245	25	37.00 (3.54)	36.06 (4.32)	48	0.25
130	143	274	40	45.54 (1.59)	45.58 (3.27)	24	0.22

Table 14: Block size at which progressive BKZ recovers the secret, for Hankel and uniform matrices with $q = 4096$ and CBD_2. Trials are per kind of matrix; p is from a two-sided Mann–Whitney test.

Lattice geometry. Without a planted secret, we reduce three lattices with BKZ-30, with $m = 1.2n$ and 32 matrices of each kind: the lattice $\{(w, \varphi) : w \equiv A^\top \varphi\}$ of the dual attack, the lattice $\{(x, y) : x \equiv Ay\}$, and $\{\varphi : A^\top \varphi \equiv 0\}$, each modulo q, at $n = 100$ and $n = 120$. In the lattice of the dual attack, the first reduced vector lies 3.442 and 3.432 bits above the Gaussian heuristic for Hankel and uniform matrices at $n = 100$ ($p = 0.67$), and 4.394 and 4.404 bits at $n = 120$ ($p = 0.56$). In the other two lattices, BKZ-30 mostly finds only the trivial vectors of length q; where it does better, at $n = 100$ in $\{(x, y) : x \equiv Ay\}$, the first vectors again agree ($p = 0.84$). The Gram–Schmidt slopes agree in all six cases, to within 1.2×10^{-4} ($p \geq 0.12$).

Algebraic decoding. XL needs a field, so we use GF(4093), the largest prime below 2^{12}, and CBD_1 secret and noise. Each sample gives the cubic $\ell_i(\ell_i - 1)(\ell_i + 1) = 0$ with $\ell_i = b_i - \langle A_i, x \rangle$, and each coordinate gives $x_j^3 - x_j = 0$. XL at degree D multiplies every equation by every monomial of degree at most $D - 3$ and succeeds when the row space of the resulting Macaulay matrix contains $x_j - s_j$ for every j. For $(n, D) = (5, 4)$, $(5, 5)$, $(6, 4)$, $(6, 5)$, $(7, 5)$, $(8, 5)$ and $(8, 6)$, with every m around the threshold and 24 instances of each kind, all instances fail below the same m and all succeed from it on: 16, 7, 24, 11, 15, 21 and 12 samples. Their Macaulay matrices have kernels of the same dimension at every point, and both behave like generic systems, succeeding at or just past the point where the Macaulay matrix has as many rows as columns. Working over GF(4093) isolates the effect of the Hankel dependencies in a field; it removes the zero divisors and the 2-adic structure of $\mathbb{Z}_{2^{12}}$.

Algebraic systems over the ring. We therefore repeat the comparison over $\mathbb{Z}_{16}$ and $\mathbb{Z}_{2^{12}}$ themselves. There the cubic $t^3 - t$ has five roots, $\{0, \pm 1, 2^{k-1} \pm 1\}$ modulo 2^k, so the system can have spurious solutions, and Gröbner bases over the ring do not isolate the secret even when it is the only solution: $8x(x - 1)$, for example, vanishes on every root modulo 16 without lying in the ideal. We therefore count the solutions exactly, testing all 5^n candidates against the m equations; any algebraic solver must at least resolve this set. For $n = 4, 6, 8$ and m around the threshold at which the solution becomes unique, with 64 instances of each kind per cell, the fraction of

uniquely solvable instances differs between Hankel and uniform systems by at most 0.125, and in 22 of 24 cells the 95% confidence interval of the difference contains zero. We repeated the two cells that did not, both at $m = 16$, with 512 fresh instances of each kind, together with the third cell at $m = 16$ and two neighbors: the differences shrink to at most 0.020, with mixed signs, every interval contains zero, and two one-sided tests bound the difference below 0.018 to 0.054. These experiments bound the effect of the Hankel structure on the solution sets of Arora–Ge systems over the scheme's ring at toy size; bit-level algebraic and SAT solvers, and larger dimensions, remain outside their scope. These experiments detect structure that lattice reduction exploits up to dimension about 270, or that changes the rank of Arora–Ge systems at toy sizes, but not attacks that appear only at larger dimensions or use neither technique.